

BIOMECHANICS (EPHE 341)

Work, energy and power

McGinnis Chpt 4

By the end of this lecture you should be able to...

- **State** how work, energy and power are defined and how the three are related
- **Apply** the work–energy theorem and conservation of mechanical energy
- **Calculate** joint power as the product of moment and angular velocity
- **Distinguish** kinetic, gravitational and elastic energy, and add them
- **Separate** positive from negative work, and generation from absorption
- **Interpret** the stretch–shortening cycle and gait in energetic terms

Two ways to describe the same motion

Nothing here is new physics — it is a change of bookkeeping

FORCE AND MOMENTUM — what you have already seen

- Newton's second law: $\Sigma F = m \cdot a$
- Impulse–momentum: $F \cdot \Delta t = \Delta(m \cdot v)$
- Force accumulated over **time**
- Answers: *how fast is it going?*

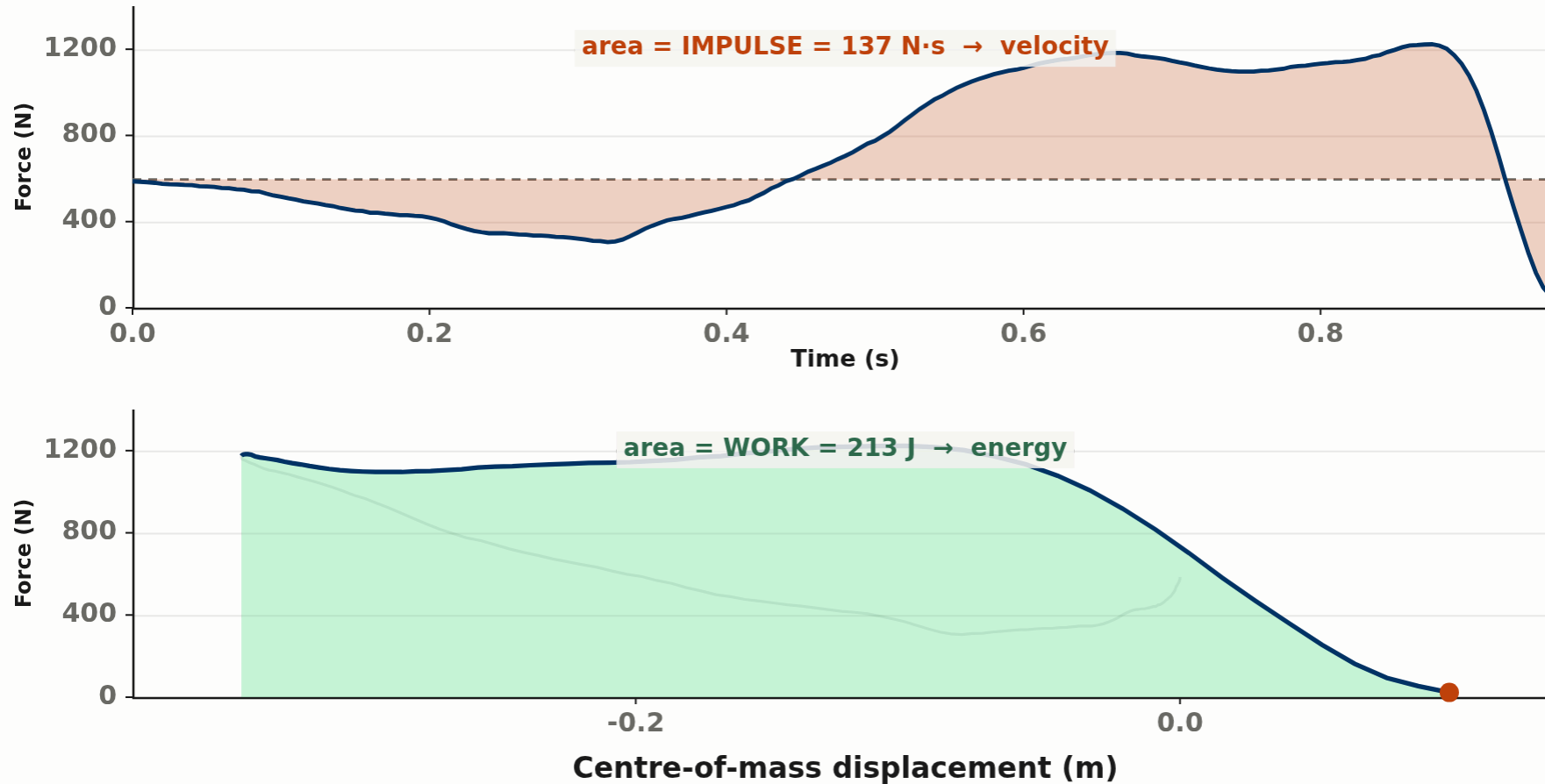
WORK AND ENERGY — what we add today

- Work: force accumulated over **distance**
- Energy: the capacity to do work
- Power: the **rate** at which work is done
- Answers: *how high, how far, how much did it cost?*

Both routes start from the same force record. Integrate it over **time** and you get velocity. Integrate it over **distance** and you get energy. Choose whichever one matches what the question gives you.

One force record, two integrals

The same measured jump, read two different ways



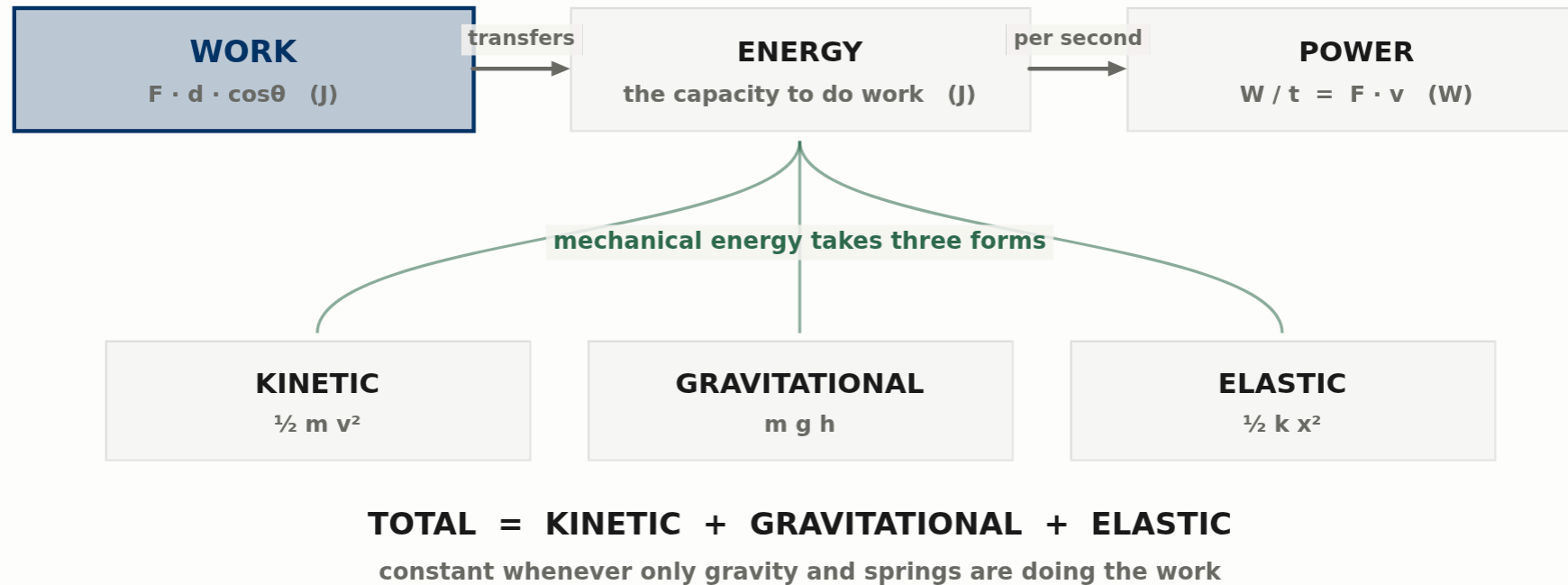
the same record, two x axes · impulse **137 N·s** gives $v = 2.25 \text{ m/s}$ · work **213 J** gives $\Delta KE + \Delta PE$ · force $\times$ time, or force $\times$ distance

Through the
jump

0.96 s

How the three fit together

The whole lecture on one slide — click any box



WORK — a force acting through a distance · $W = F \cdot d \cdot \cos\theta$, in joules. Work is how energy gets moved from one place to another — it is a transfer, not a thing a body has. Positive work puts energy in, negative work takes it out.

work

energy

power

kinetic

gravitational

elastic

A force acting through a displacement

$$W = F \cdot d \cdot \cos \theta$$

SI unit

joule (J)

1 J = 1 N·m

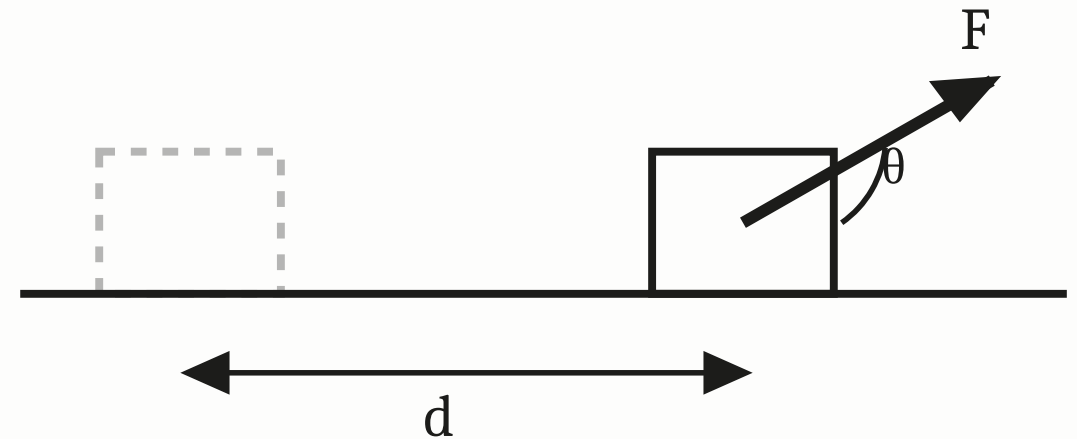
A scalar

No direction of its own. The sign comes entirely from $\cos \theta$.

θ

The angle between the force and the **displacement** — not the ground.

No displacement means no work. Holding a barbell overhead is exhausting and mechanically zero.

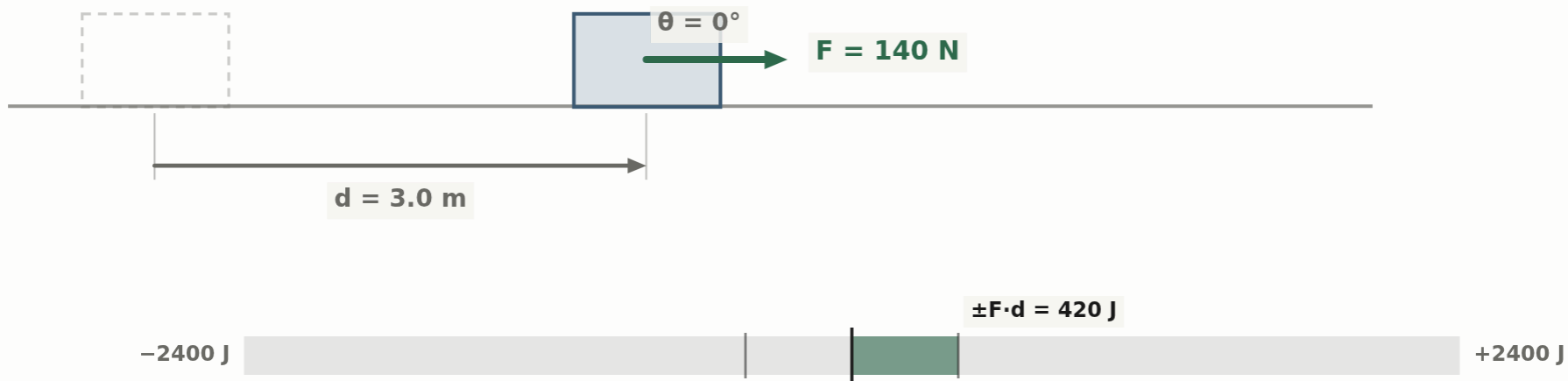


Only the component of F along d does work

The sign of work

All three of F , d and θ are drawn to scale — move any of them

POSITIVE WORK



$$W = F \cdot d \cdot \cos\theta = 140 \times 3.0 \times \cos 0^\circ = \mathbf{420 \text{ J}} \cdot \cos\theta = 1.00$$

Angle θ

Distance

Force

Positive and negative work

The muscle is doing both, every step you take

POSITIVE WORK —

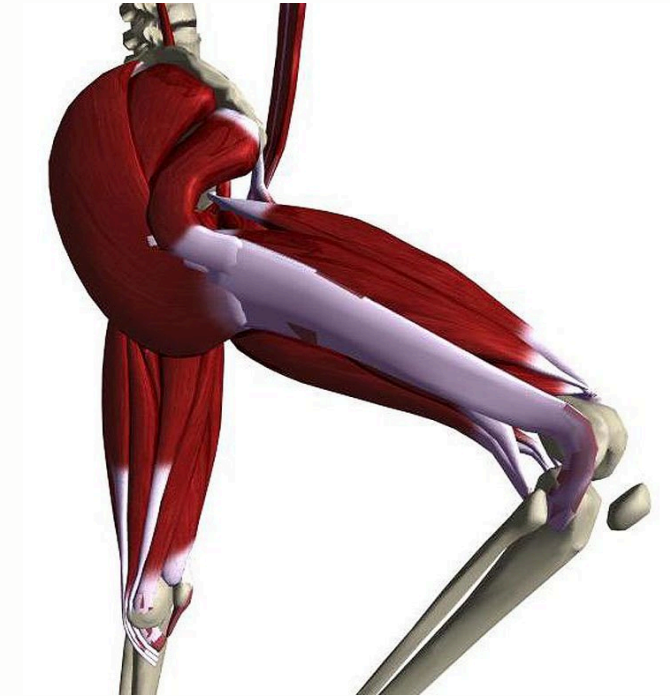
$$\theta < 90^\circ$$

- Force and motion in the same direction
- The muscle **shortens** under load — **concentric**
- Energy flows **into** the body segment
- Pushing out of the bottom of a squat

NEGATIVE WORK —

$$\theta > 90^\circ$$

- Force opposes the motion
- The muscle **lengthens** under load — **eccentric**
- Energy flows **out of** the segment
- Lowering into a squat, or landing



Quadriceps: concentric on the way up, eccentric on the way down

Negative work is not wasted work. It is how you stop, how you land, and — in a countermovement — how

Not all of the work a body does moves the body

EXTERNAL WORK

- Work done on objects or against the environment
- Raising the centre of mass, lifting a load, overcoming drag
- What a force plate or an ergometer measures

INTERNAL WORK

- Work done moving the segments relative to the centre of mass
- Swinging the legs and arms in running
- Invisible to the force plate, but it still costs energy

A runner on a treadmill at constant speed raises their centre of mass by nothing over a full stride, so the external work is near zero — and yet they are clearly working hard. All of it is internal.

The capacity to do work — and it comes in three mechanical forms

$$E_{\text{mech}} = KE + U_g + U_e$$

KINETIC

$$KE = \frac{1}{2} \cdot m \cdot v^2$$

Because it is moving

GRAVITATIONAL

$$U_g = m \cdot g \cdot h$$

Because of where it is

ELASTIC

$$U_e = \frac{1}{2} \cdot k \cdot x^2$$

Because something is deformed

- All three are **scalars**, measured in joules — the same unit as work
- They simply **add**; there is no vector sum to do
- Work is how energy **moves** between forms and between bodies
- Energy is never created here, only shifted or dissipated

Kinetic energy

The energy an object has because it is moving

$$KE = \frac{1}{2} \cdot m \cdot v^2$$

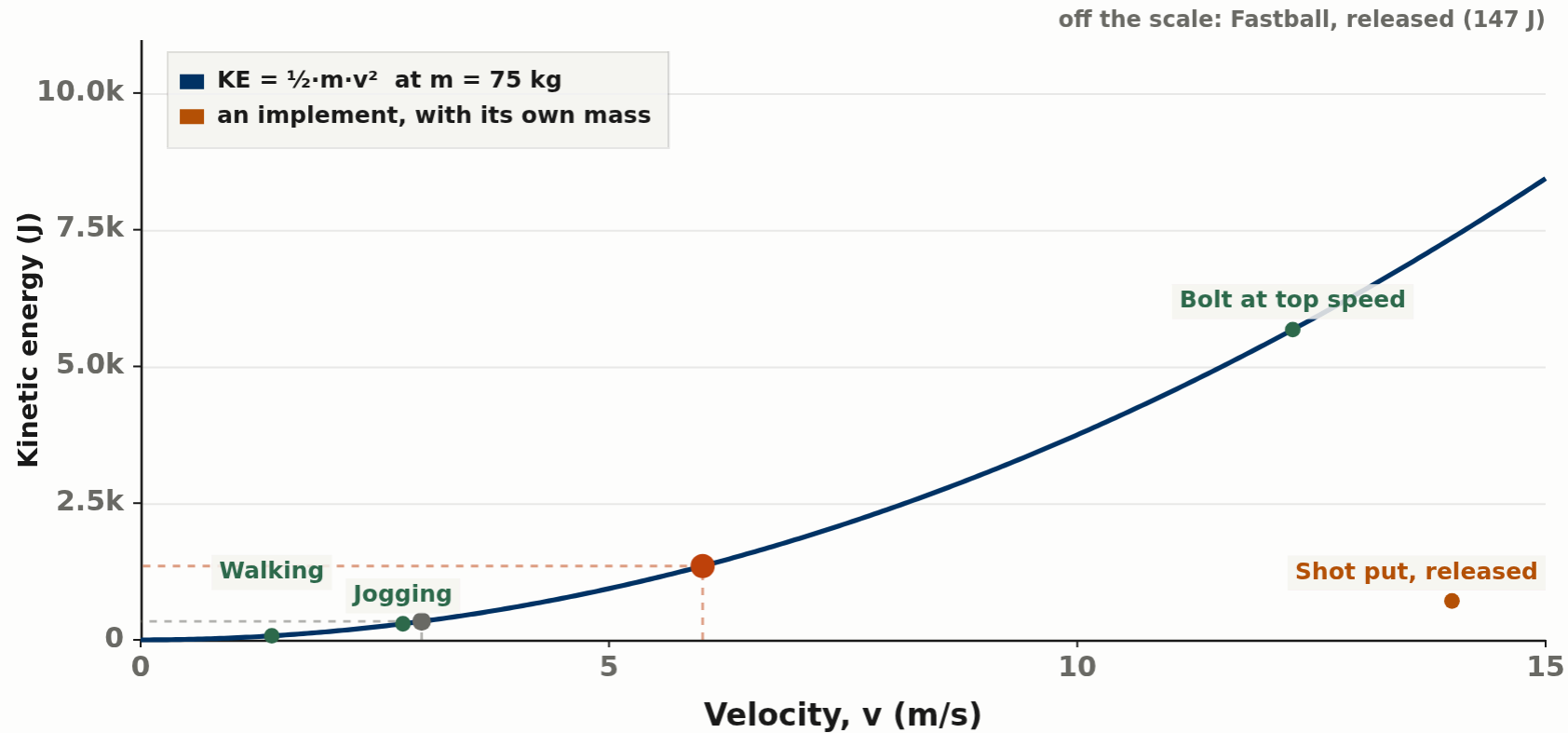
- A scalar, and always ≥ 0 — there is no such thing as negative kinetic energy
- Mass enters **once**; velocity enters **twice**
- **Doubling the speed quadruples the energy.** Halving the mass only halves it
- This asymmetry is why injury severity scales so brutally with speed



70 kg at 10 m/s carries 3,500 J

Kinetic energy

Doubling the velocity quadruples the energy



at **3.0 m/s** → 338 J · at **6.0 m/s** → **1350 J** — twice the speed, 4.0× the energy

set the speed:

walking

jogging

sprinting

Momentum and energy are not the same thing

Double the speed: momentum doubles, kinetic energy quadruples

	V = 5 M/S	V = 10 M/S	CHANGE
Momentum $p = m \cdot v$	350 kg·m/s	700 kg·m/s	×2
Kinetic energy $\frac{1}{2}mv^2$	875 J	3500 J	×4

- Momentum tells you what it takes to **stop it in a given time**
- Kinetic energy tells you what it takes to **stop it in a given distance**
- A 70 kg athlete at 10 m/s and a 140 kg athlete at 5 m/s have the same momentum — and the lighter, faster one carries **twice** the energy



Both quantities are conserved in a collision — but they are not interchangeable

Net work equals the change in kinetic energy

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

WHAT IT SAYS

- $W_{\text{net}} > 0 \rightarrow$ the object speeds up
- $W_{\text{net}} < 0 \rightarrow$ it slows down
- $W_{\text{net}} = 0 \rightarrow$ the speed is unchanged

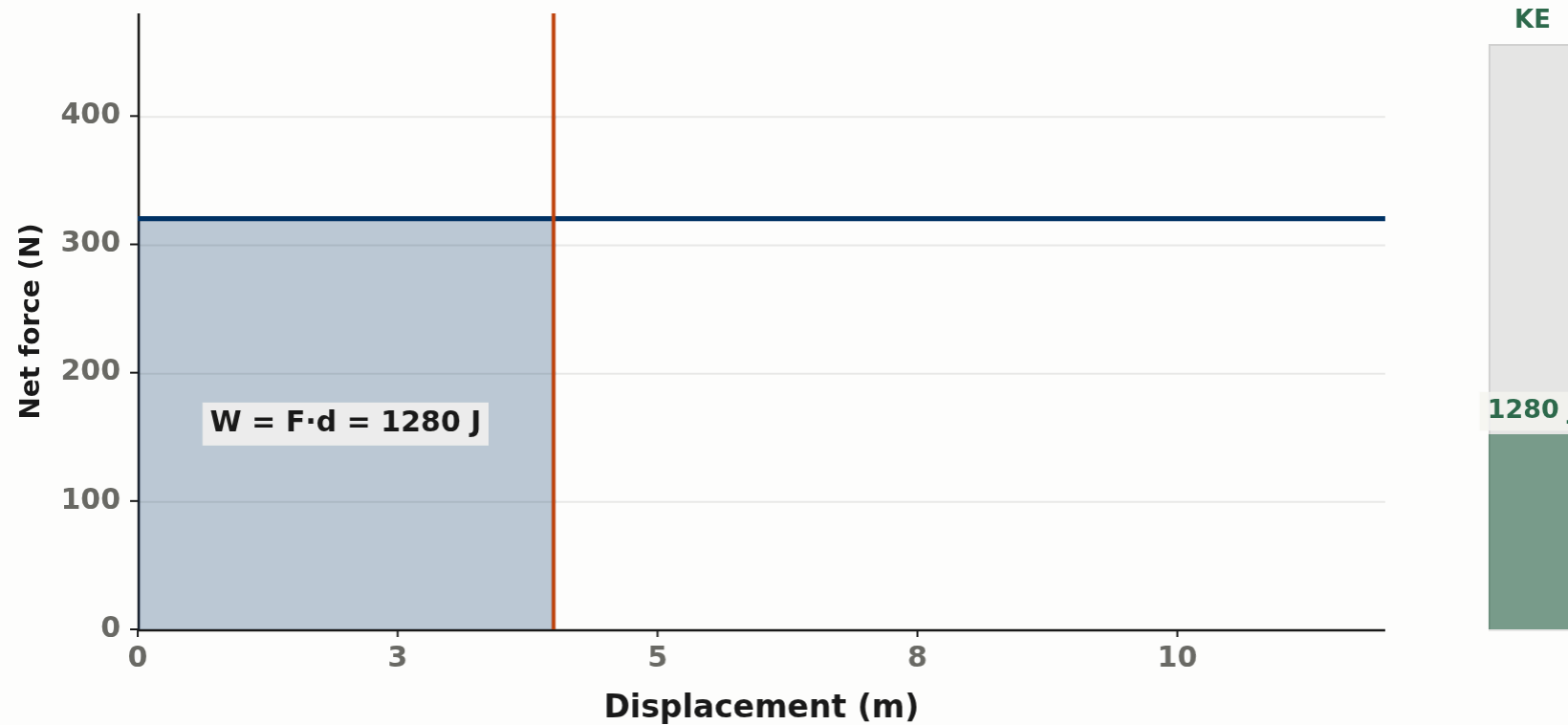
WHERE IT IS USED

- Sprint acceleration: net forward work becomes kinetic energy
- Braking at heel-strike: negative work removes it again
- Throwing: work on the implement becomes its kinetic energy

It is a **theorem**, not a definition — it follows from $\Sigma F = ma$. And it is **net** work: every force that acts counts, friction and gravity included.

The work–energy theorem

The area and the energy are the same number



$$W_{\text{net}} = 1280 \text{ J} \cdot \Delta KE = \frac{1}{2}(75)(5.84)^2 = 1280 \text{ J} \cdot v = 5.84 \text{ m/s after 4.0 m}$$

▶ Play

Distance
run

4.0 m

Net force

320 N

Mass

75 kg

Gravitational potential energy

Energy stored in an object's vertical position

$$U_g = m \cdot g \cdot h$$

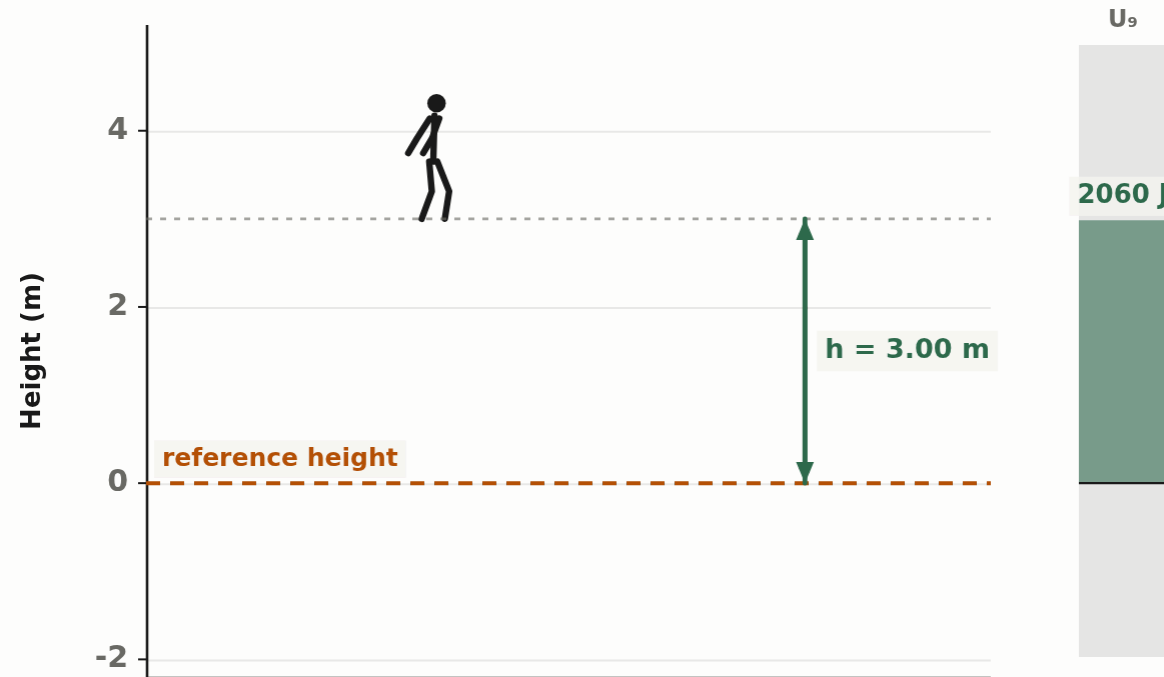
- **h is measured from a reference height that you choose** — only *changes* in U_g are physically meaningful
- Raising 1 kg by 1 m stores about 9.81 J
- A change in U_g is minus the work done by gravity
- Linear in height — unlike the other two, nothing is squared



Every metre gained is $m \cdot g$ joules banked

Gravitational potential energy

Move the reference and the number changes; the climb does not



$$U_g = m \cdot g \cdot h = 70 \times 9.81 \times 3.00 = \mathbf{2060 \text{ J}}$$

· move the reference and U changes; the climb does not

Mass Height
Reference

Elastic potential energy

Energy stored in a deformed spring — or a tendon

$$U_e = \frac{1}{2} \cdot k \cdot x^2$$

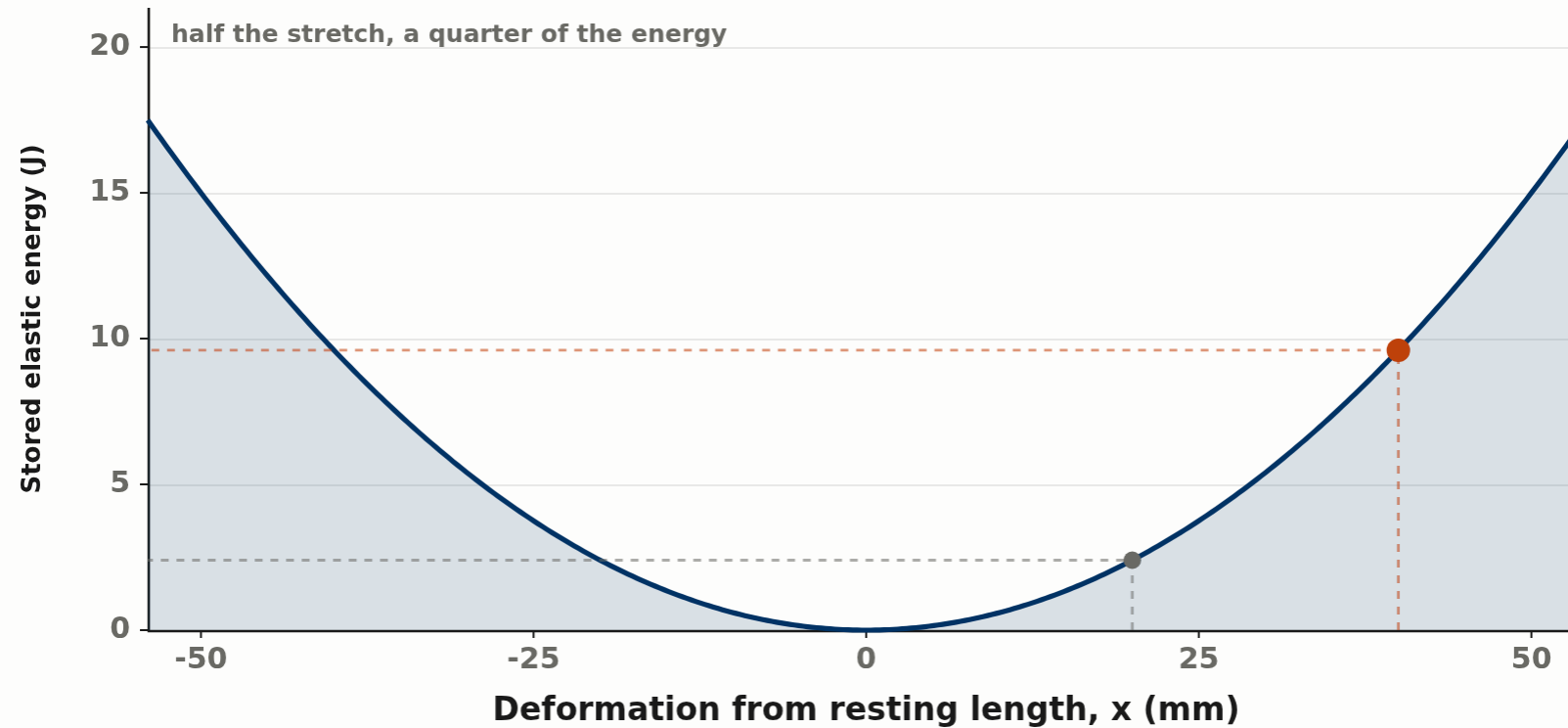
- k is the stiffness in N/m, x the deformation from resting length
- **Double the stretch, four times the energy** — same quadratic structure as KE
- Tendons, ligaments and aponeuroses are all biological springs
- Real tissue returns about 90–93% of what it stores; the rest is lost as heat



A horse's distal limb is almost all tendon and bone — spring, not motor

Elastic potential energy

Stiffness and stretch do not contribute equally



$$U = \frac{1}{2} \cdot k \cdot x^2 = \frac{1}{2} \times 12.0 \text{ N/mm} \times (40 \text{ mm})^2 = \mathbf{9.60 \text{ J}}$$

· at 20 mm it is only 2.40 J

elastic band

Achilles tendon

vaulting pole

Stiffness k



12.0
N/mm

Stretch x



40 mm

Conservation of mechanical energy

When only gravity and springs act, the total does not change

$$KE + U_g + U_e = \text{constant}$$

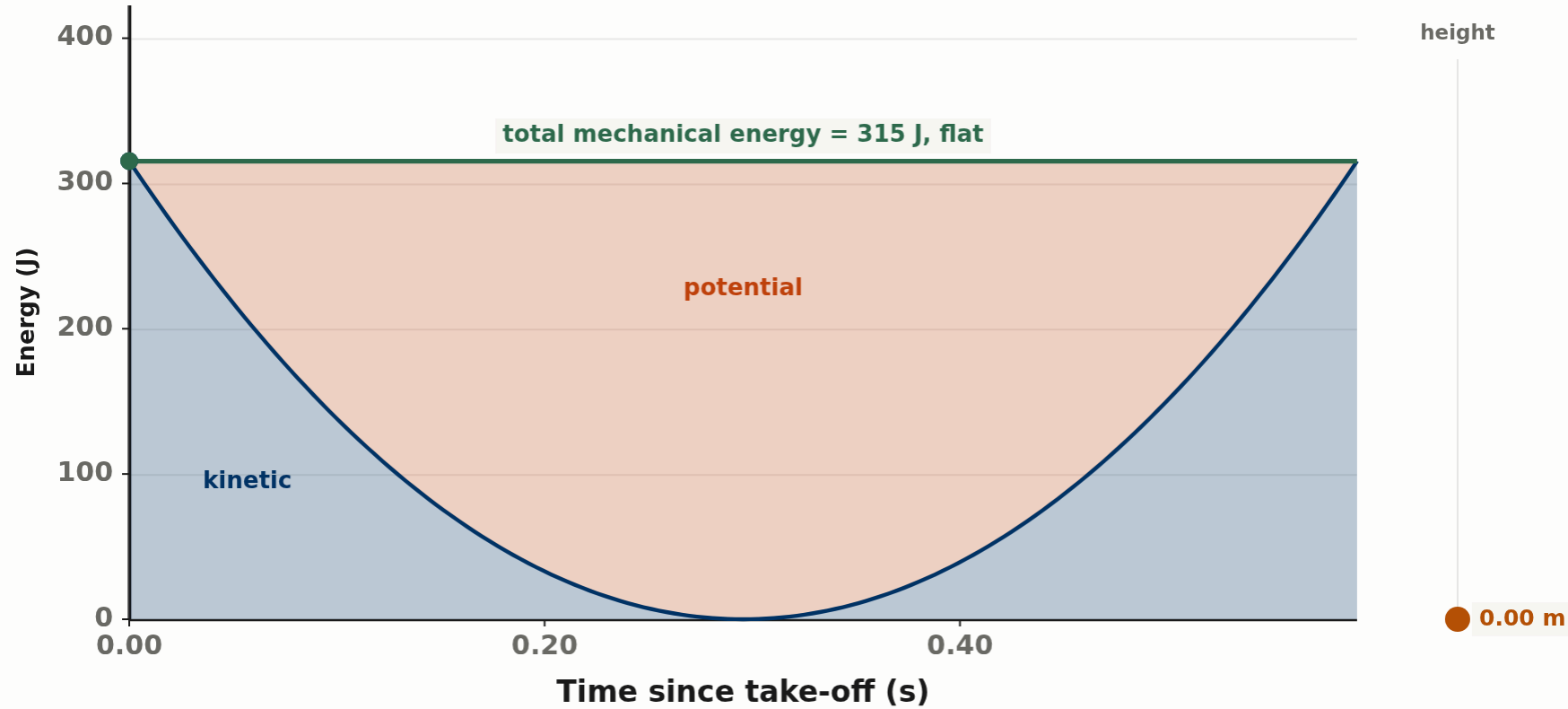
- The three terms trade back and forth, but their sum is flat
- This holds **only** without friction, drag or muscle work
- In the **flight phase** of a jump it is exactly true — nothing but gravity acts
- During push-off it fails completely, because the muscles are pouring energy in



Run-up KE → pole U_e → height U_g , all three in one movement

Conservation of mechanical energy

Kinetic into potential and back again, through one flight



KE 315 J + PE 0 J = 315 J · peak rise $\Delta h = v^2/2g = 0.429$ m, whatever the mass

▶ Play

Through the
flight



0.00 s

The rate at which work is done, or energy transferred

$$P = W / t$$

Time-average
power

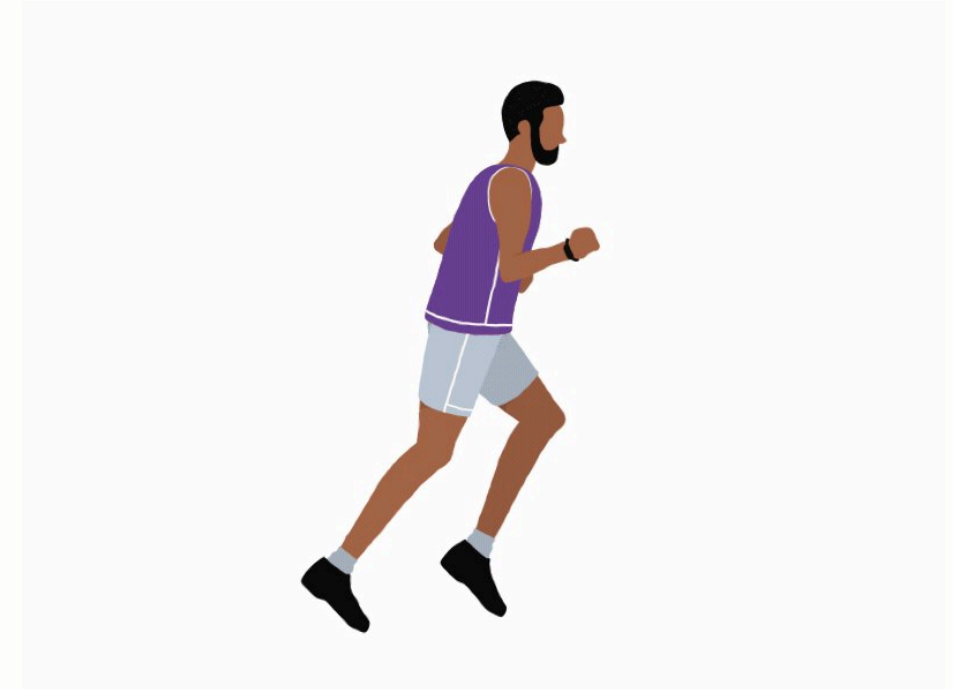
$$P = F \cdot v$$

Instantaneous
linear power

$$P = M \cdot \omega$$

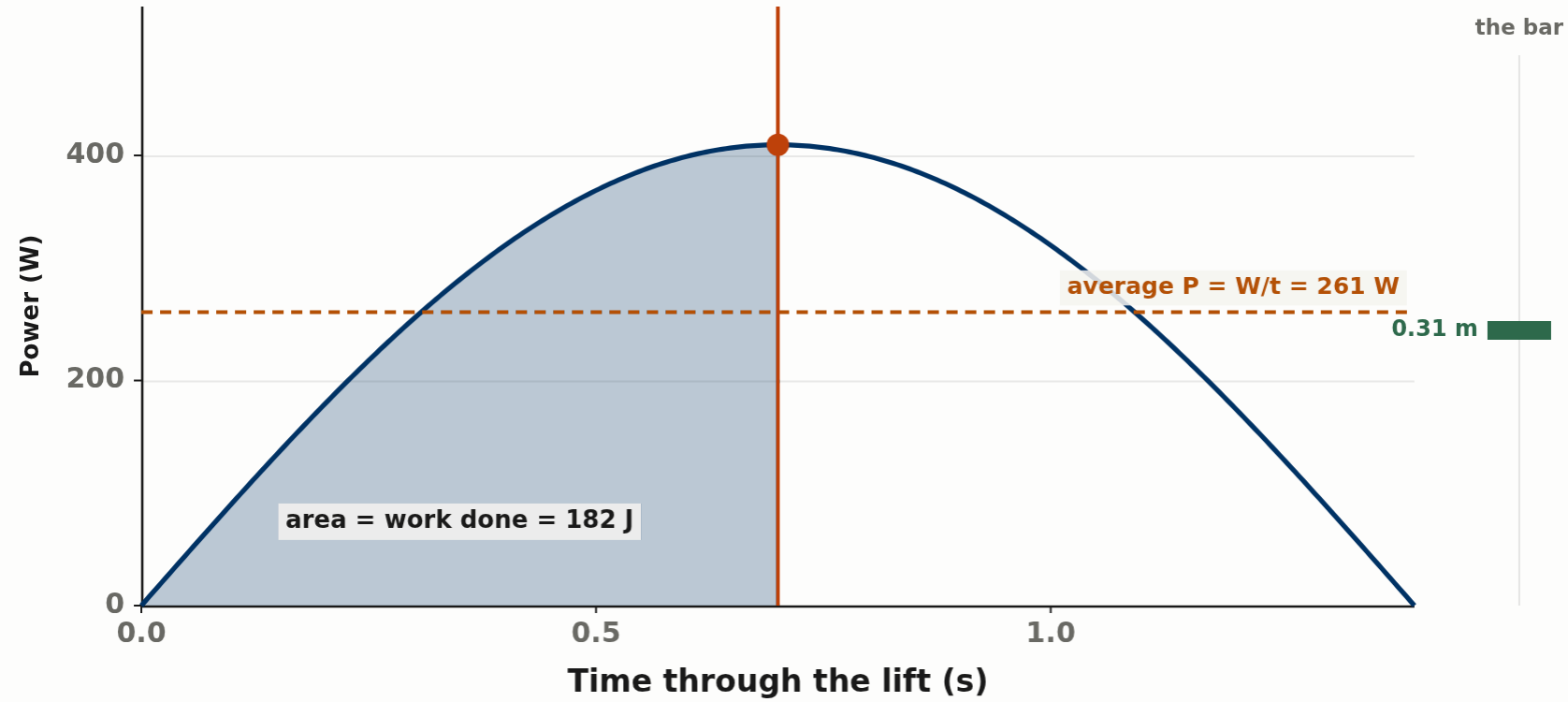
Joint power —
moment ×
angular velocity

- SI unit: the **watt** — 1 W = 1 joule per second
- These are one quantity written three ways, not three quantities
- $P = F \cdot v$ follows from $P = W/t$: $W = F \cdot d$, divide by t , and d/t is v
- 1 horsepower $\approx$ 746 W; a fit cyclist holds about 300 W for an hour



Same work, half the time, double the power

One lift, three formulas, the same watts



$$P = F \cdot v = 589 \text{ N} \times 0.70 \text{ m/s} = 409 \text{ W} \cdot W/t = 182 \text{ J} / 0.70 \text{ s} = 261 \text{ W}$$

$P = F \cdot v$

$P = \tau \cdot \omega$

Through the lift

0.70 s

Load

60 kg

Lift takes

1.40 s

The framework, in one table

Everything from here on is an application of these four lines

	DEFINITION	UNIT	WHAT IT TELLS YOU
Work	$W = F \cdot d \cdot \cos \theta$	J	How much energy was transferred by a force acting through a distance
Energy	$KE + U_g + U_e$	J	How much capacity to do work a body holds , in three forms that add
Work–energy	$W_{\text{net}} = \Delta KE$	J	Net work is the change in kinetic energy
Power	$P = W/t = F \cdot v = M \cdot \omega$	W	How fast that transfer happens

Energy is the currency, work is the transaction, power is the rate. Every problem in the next section is asking where a fixed number of joules went.

Part 02

Working problems

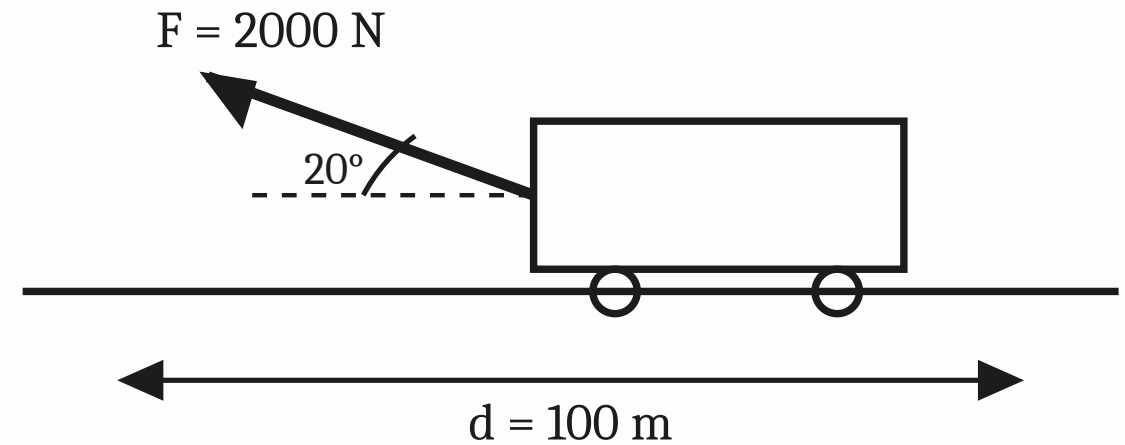
Eight of them, in the order the framework was built

Problem 1 — work at an angle

$$W = F \cdot d \cdot \cos \theta$$



A strongman competitor pulls a truck 100 m along a flat road. The harness rope runs up to the truck at **20° above horizontal**, and he maintains a steady **2000 N** of tension. How much work does he do on the truck?



Given:

$$F = 2000 \text{ N}$$

$$d = 100 \text{ m (horizontal)}$$

$$\theta = 20^\circ$$

Problem 1 — solution

Only the component along the motion counts

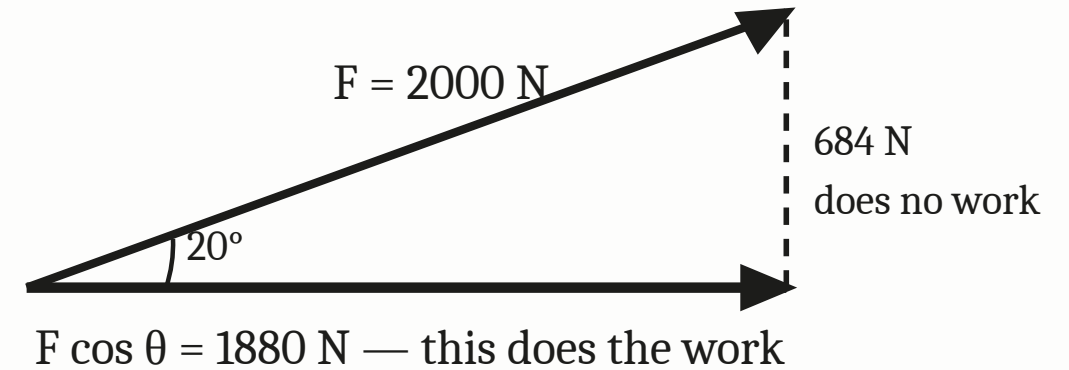
$$W = F \cdot d \cdot \cos \theta$$

$$W = (2000 \text{ N})(100 \text{ m})(\cos 20^\circ)$$

$$W = (2000)(100)(0.940)$$

$$W \approx 188,000 \text{ J} = 188 \text{ kJ}$$

- At 0° — rope horizontal — it would have been the full 200 kJ
- The missing 12 kJ was never lost; that component was lifting, not pulling, and the truck did not move upward
- The **vertical** component, $2000 \cdot \sin 20^\circ = 684 \text{ N}$, does **zero** work over the whole 100 m



Resolve the force; keep the component along d

Problem 2 — work against gravity

A lift at constant velocity

An athlete lifts a **100 kg** barbell **2.1 m** from the floor to overhead at a roughly constant velocity.

How much work does the athlete do on the bar?

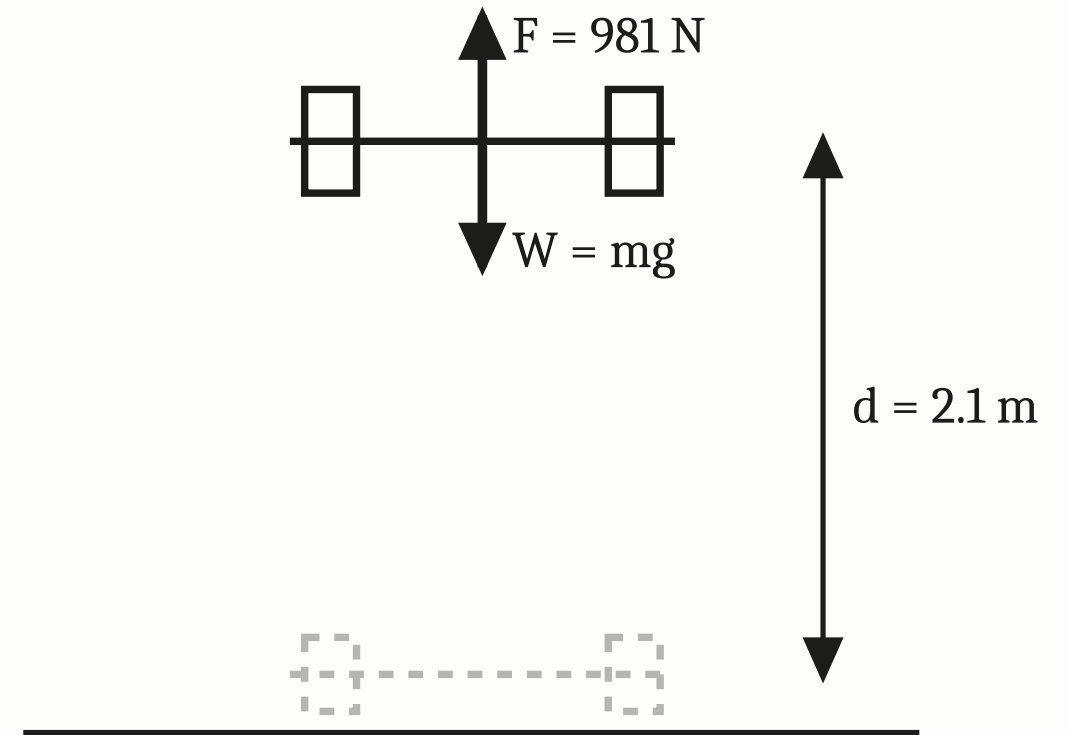
Given:

$$m = 100 \text{ kg}$$

$$d = 2.1 \text{ m (vertical)}$$

$$a \approx 0 \text{ (constant velocity)}$$

$$\theta = 0^\circ \text{ (force and displacement both up)}$$



At constant velocity the two forces are equal and opposite

Problem 2 — solution

And the first appearance of a conversion

- At constant velocity $\Sigma F = 0$, so the lifting force equals the weight of the bar

$$F = m \cdot g = (100 \text{ kg})(9.81 \text{ m/s}^2) = 981 \text{ N}$$

- The force and the displacement are both upward, so $\theta = 0^\circ$ and $\cos \theta = 1$

$$W = F \cdot d \cdot \cos(0^\circ) = (981 \text{ N})(2.1 \text{ m})(1)$$

$$W \approx 2060 \text{ J}$$

Those 2060 J did not vanish — they are sitting in the bar's new position as gravitational potential energy, and you can get them back by dropping it. **$W_{\text{against gravity}} = \Delta U_g$** .

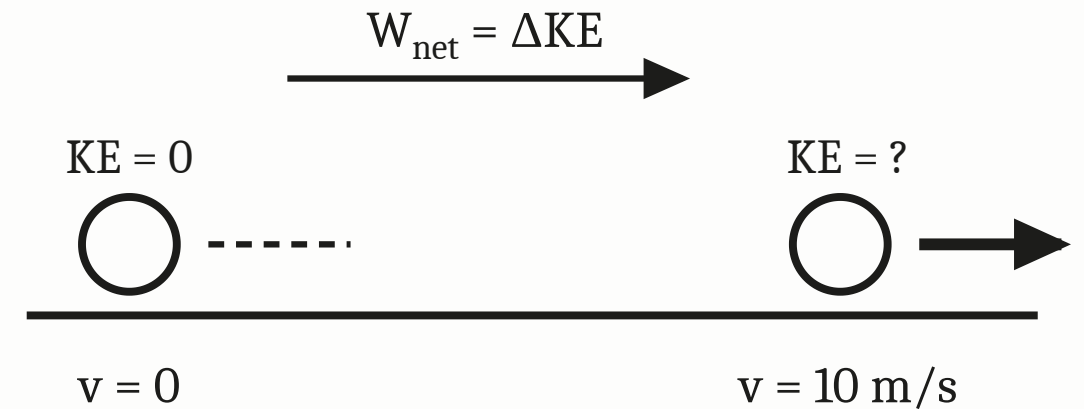
Problem 3 — kinetic energy

And what happens when you change the speed



A **70 kg** sprinter is running at **10 m/s**.

- (a) How much kinetic energy do they have?
- (b) How much work was required to get them there from a standing start?
- (c) What happens to (a) if they reach 12 m/s?



Given:

$$m = 70 \text{ kg}$$

$$v_i = 0, v_f = 10 \text{ m/s}$$

$$\text{then } v_f = 12 \text{ m/s}$$

Problem 3 — solution

A 20% increase in speed costs 44% more energy

(a) at 10 m/s

$$KE = \frac{1}{2} \cdot m \cdot v^2 = \frac{1}{2}(70)(10)^2$$

$$KE = 3500 \text{ J}$$

(b) starting from rest, $v_i = 0$ so ΔKE is all of it

$$W_{\text{net}} = \Delta KE = 3500 - 0 = \mathbf{3500 \text{ J}}$$

(c) at 12 m/s

$$KE = \frac{1}{2}(70)(12)^2 = 5040 \text{ J}$$

an extra **1540 J** — 44% more

V (M/S)	KE (J)	RELATIVE
5	875	0.25 ×
10	3500	1.00 ×
12	5040	1.44 ×
20	14000	4.00 ×

The last 2 m/s cost almost half as much again as the first 10 m/s put together. That is the squared term, and it is why the final metres of a sprint are so expensive to buy.

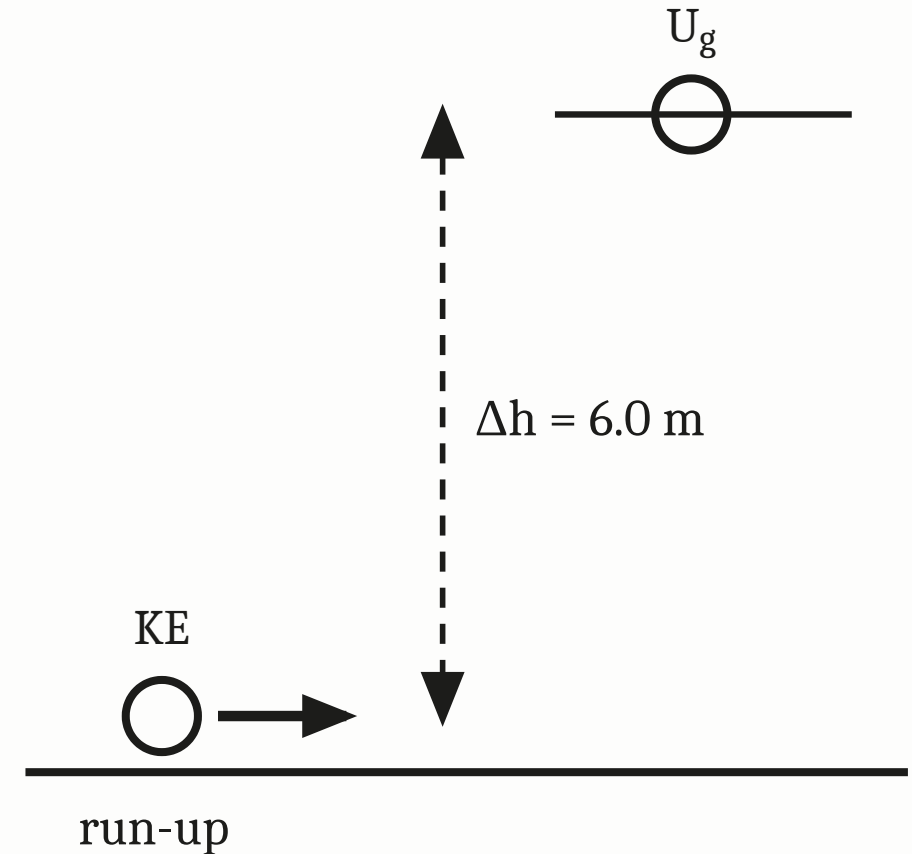
Problem 4 — gravitational potential energy

Where does a vaulter's height come from?



A **70 kg** pole vaulter raises their centre of mass to **6.0 m** above its standing height.

- (a) How much gravitational potential energy have they gained?
- (b) How fast would they have to be running for their kinetic energy alone to supply it?



Given:

Problem 4 — solution

And why vaulting is not purely a conversion

(a)

$$\Delta U_g = m \cdot g \cdot \Delta h = (70)(9.81)(6.0)$$

$$\Delta U_g \approx \mathbf{4120 \text{ J}}$$

(b) set the kinetic energy equal to it — the mass cancels

$$\frac{1}{2} \cdot m \cdot v^2 = m \cdot g \cdot \Delta h$$

$$v = \sqrt{(2 \cdot g \cdot \Delta h)} = \sqrt{(2 \times 9.81 \times 6.0)}$$

$$v \approx \mathbf{10.8 \text{ m/s}}$$

10.8 m/s is faster than almost any vaulter runs — elite approach speeds are about 9.5 m/s, worth roughly 3160 J.

The missing **960 J** is muscular work done during the plant, the swing and the pull. The pole is a very good spring, but it is not a free lunch.

- Note the **mass cancelled** in (b) — this is the same cancellation that appears in every jump-height calculation
- Nothing here depended on the shape of the vault, only on its two endpoints

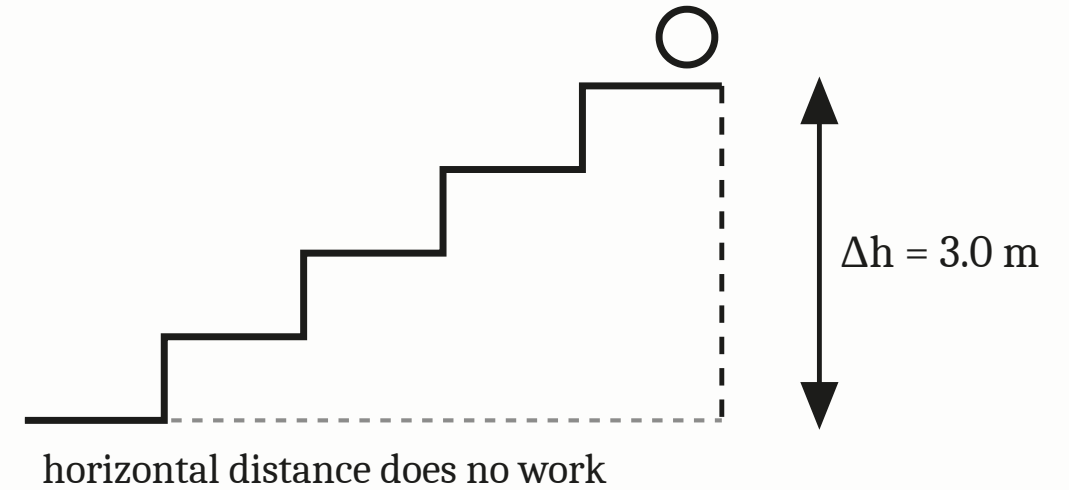
Problem 5 — work and power together

The same climb, asked two ways



A **70 kg** student climbs a staircase **3.0 m** high.

- (a) How much gravitational potential energy do they gain?
- (b) If the climb takes **2.0 s**, what average power did they produce?
- (c) And if they walk it in 8.0 s instead?



Given:

$$m = 70 \text{ kg}$$

$$\Delta h = 3.0 \text{ m}$$

$$\Delta t = 2.0 \text{ s, then } 8.0 \text{ s}$$

Problem 5 — solution

Same work, very different power

$$\Delta U_g = m \cdot g \cdot \Delta h = (70)(9.81)(3.0)$$

$$\Delta U_g \approx \mathbf{2060 \text{ J}}$$

$$P = W / t = 2060 \text{ J} / 2.0 \text{ s}$$

$$P \approx \mathbf{1030 \text{ W}} \text{ (about 1.4 hp)}$$

$$P = 2060 \text{ J} / 8.0 \text{ s}$$

$$P \approx \mathbf{258 \text{ W}}$$

2060 J is the same number as the barbell in Problem 2 — a 100 kg bar raised 2.1 m and a 70 kg student raised 3.0 m are the same job.

Once it is energy, it does not remember how it was earned.

- 1030 W for two seconds is genuinely large — roughly three times what a trained cyclist can hold for an hour
- 258 W, walked, is about what that same cyclist *can* hold for an hour
- The difference between those two numbers is the subject of the power–duration curve in Part 03

Problem 6 — the Margaria–Kalamen stair sprint

A field test of peak anaerobic power — you will run it in the lab

A **75 kg** athlete sprints up a staircase. Timing gates are set on the **3rd** and **9th** steps, each step is **0.175 m** high, and the athlete covers the six steps between the gates in **0.52 s**.

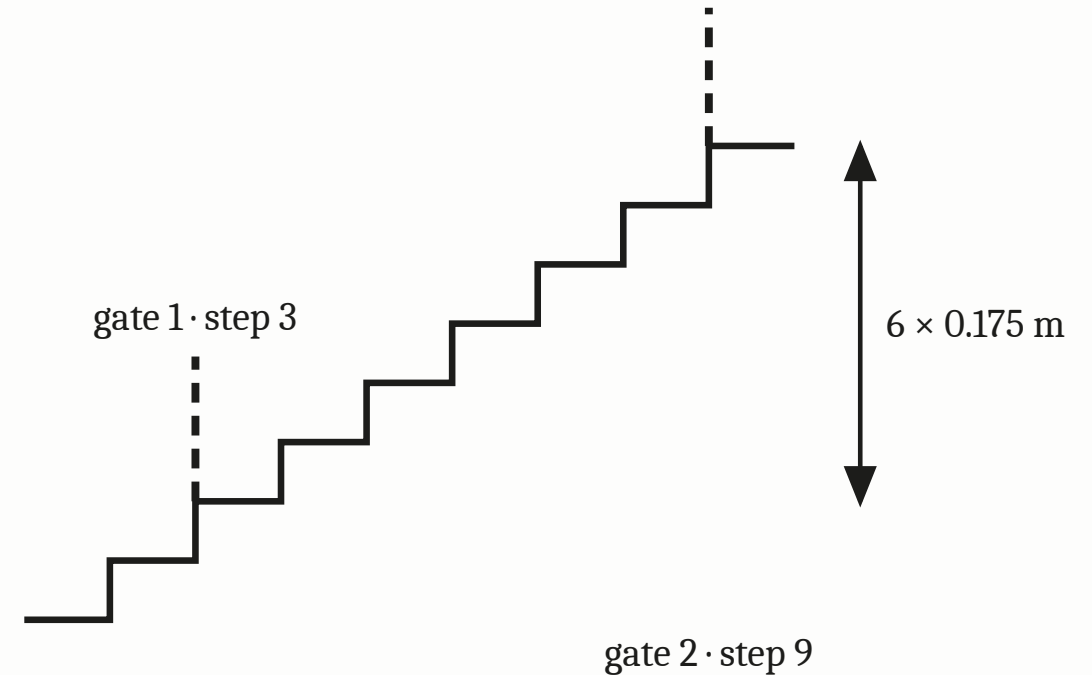
What is their peak anaerobic power, in watts and in watts per kilogram?

Given:

$$m = 75 \text{ kg}$$

$$\text{step} = 0.175 \text{ m, 6 steps between gates}$$

$$\Delta t = 0.52 \text{ s}$$



The first two steps are run-up — the clock starts at speed

Problem 6 — solution

Only the vertical rise counts

- The work being measured is against gravity, so the horizontal travel is irrelevant

$$\Delta h = 6 \times 0.175 \text{ m} = 1.05 \text{ m}$$

$$P = m \cdot g \cdot \Delta h / \Delta t$$

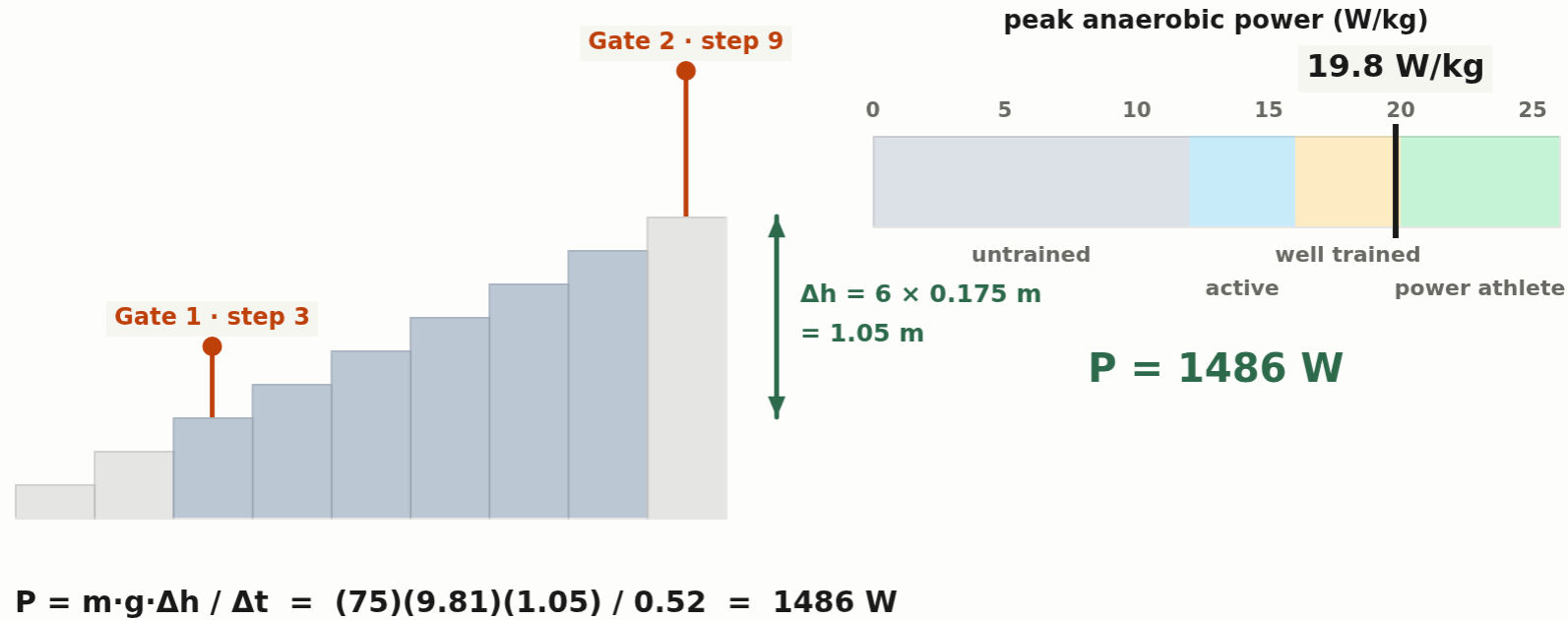
$$P = (75)(9.81)(1.05) / 0.52$$

$$P \approx \mathbf{1486 \text{ W}}$$

$$P / m = 1486 / 75 \approx \mathbf{19.8 \text{ W/kg}} \text{ — competitive for a power athlete}$$

Problem 6 — the same test, with your own numbers

Change the mass and watch what does not move



$$P = 1486 \text{ W} \cdot 19.8 \text{ W/kg} \cdot \Delta h = 1.05 \text{ m in } 0.52 \text{ s}$$

Mass

75 kg

Step
height

0.175
m

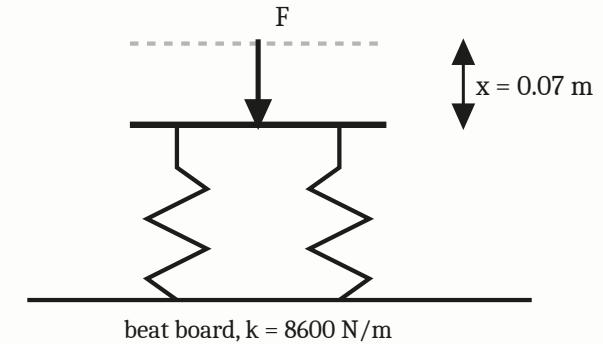
Problem 7 — elastic potential energy

A gymnast on a beat board



A gymnast lands on a beat board of stiffness $k = 8600$ **N/m** and compresses it **0.07 m**.

- (a) How much elastic energy is stored in the board?
- (b) If all of it were returned to a **55 kg** gymnast, how much extra height would it buy?



Given:

$$k = 8600 \text{ N/m}$$

$$x = 0.07 \text{ m}$$

$$m = 55 \text{ kg}$$

Problem 7 — solution

A small number, and that is the point

(a)

$$U_e = \frac{1}{2} \cdot k \cdot x^2 = \frac{1}{2}(8600)(0.07)^2$$

$$U_e = \frac{1}{2}(8600)(0.0049)$$

$$U_e \approx \mathbf{21 \text{ J}}$$

(b) set the returned elastic energy equal to a gain in height

$$U_e = m \cdot g \cdot \Delta h$$

$$\Delta h = 21 / (55 \times 9.81)$$

$$\Delta h \approx \mathbf{0.039 \text{ m}} \text{ (about 4 cm)}$$

Compress the board **twice** as far and you store **four** times as much — 84 J, worth about 16 cm. The squared term is why a heavier, faster approach pays off so disproportionately.

- 21 J is genuinely small next to the $\approx 1500 \text{ J}$ of a gymnast's run-up kinetic energy
- No real board returns 100% — hysteresis takes a cut, so 4 cm is an upper bound
- The board's main job is to **reshape the force–time curve**, letting the gymnast apply a large force for longer than a rigid floor would allow

Problem 8 — conservation, and the mass cancelling

The single most useful result in jump testing

A **75 kg** athlete leaves the ground with a vertical velocity of **2.9 m/s**. How high does their centre of mass rise?

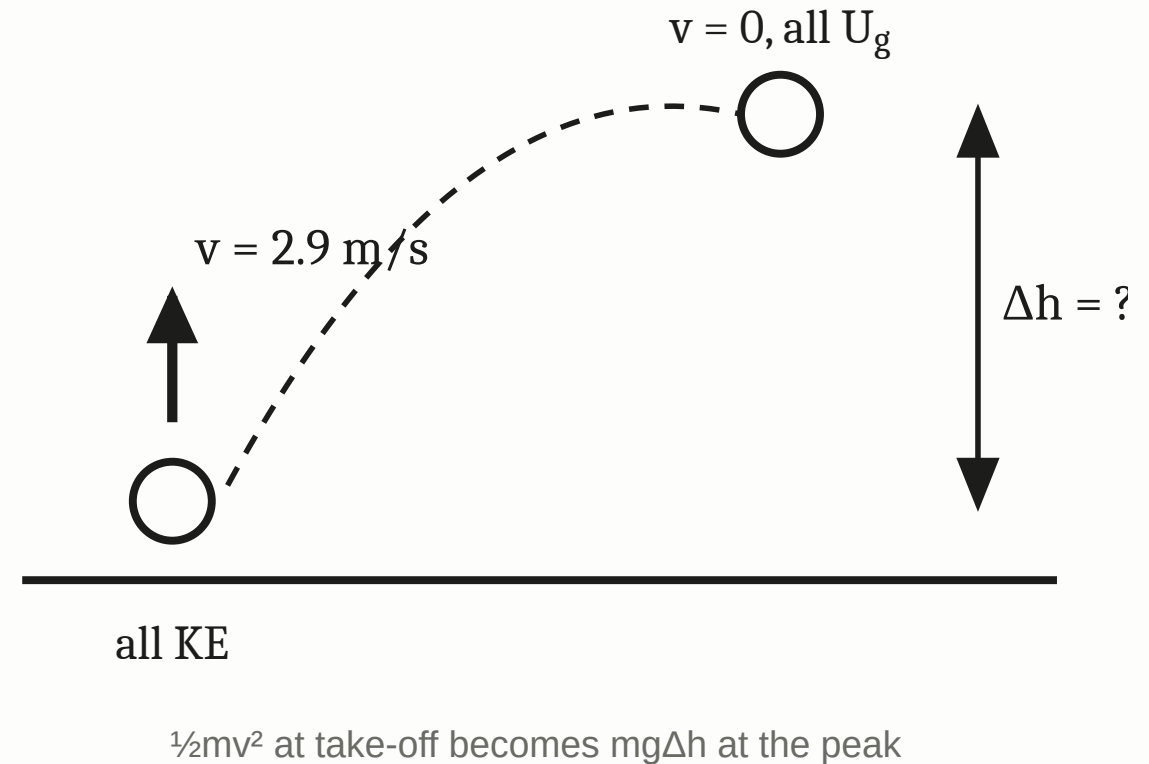
Once they are airborne, only gravity acts — so mechanical energy is conserved.

Given:

$$m = 75 \text{ kg}$$

$$v_{\text{take-off}} = 2.9 \text{ m/s}$$

$$\text{at the peak, } v = 0$$



Problem 8 — solution

The mass cancels — jump height depends only on take-off velocity

- At the peak all of the take-off kinetic energy has become potential energy

$$\frac{1}{2} \cdot m \cdot v_{\text{to}}^2 = m \cdot g \cdot \Delta h$$

$$\Delta h = v_{\text{to}}^2 / (2 \cdot g)$$

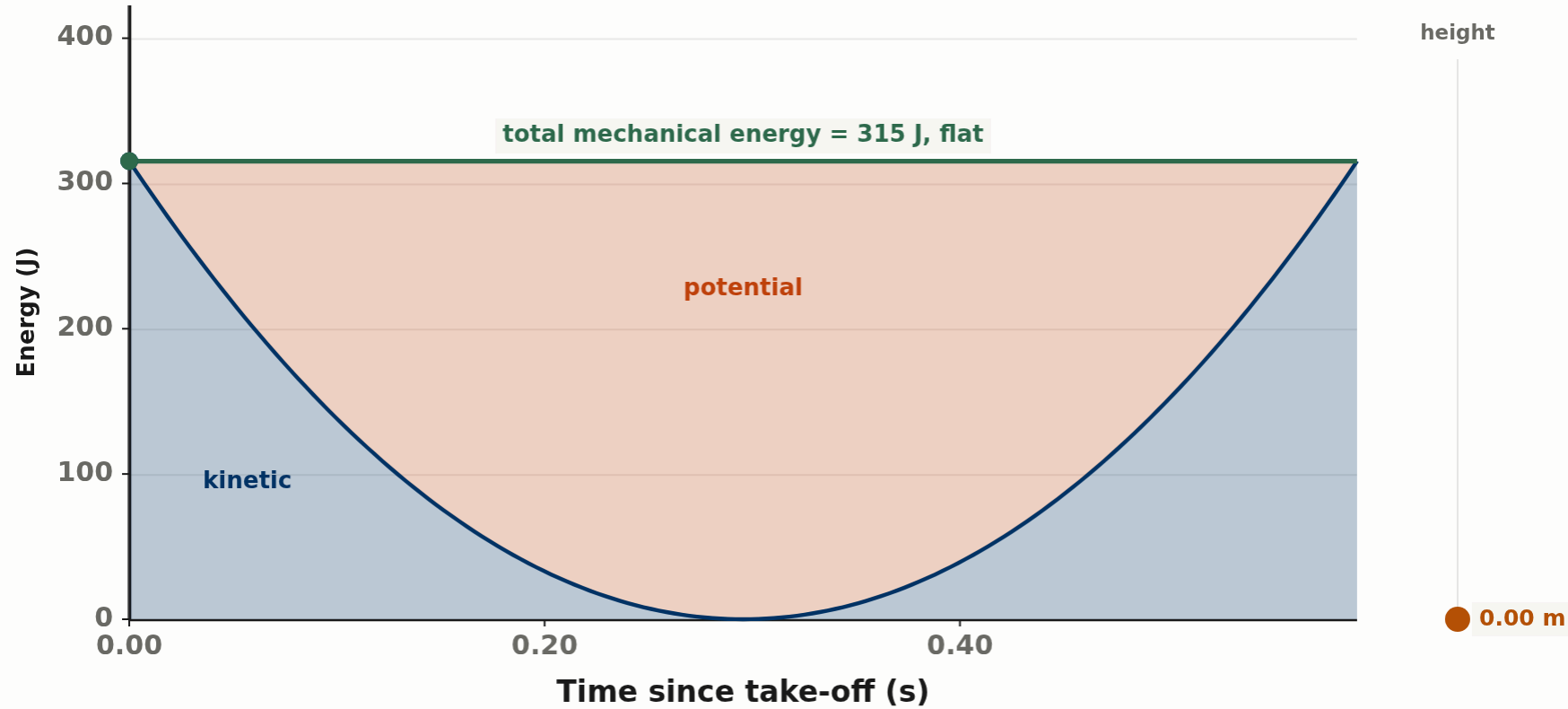
$$\Delta h = (2.9 \text{ m/s})^2 / (2 \times 9.81 \text{ m/s}^2)$$

$$\Delta h \approx \mathbf{0.43 \text{ m}} \text{ (43 cm)}$$

The mass divided out of both sides. Measure take-off velocity — or flight time — and you have the height without weighing anybody. This is why jump mats work.

Problem 8 — try to make the mass matter

Drag it from 45 kg to 110 kg and watch the height



KE **315 J** + PE **0 J** = **315 J** · peak rise $\Delta h = v^2/2g = 0.429$ m, whatever the mass

Take-off speed m/s

Mass kg

Problem 9 — energy dissipation and the helmet

The energy is fixed; all you can change is the distance

A cyclist riding at **6 m/s** falls. Their head — about **8.1%** of a 70 kg body mass, so **5.7 kg** — starts **1.5 m** above the ground and is brought to rest by the impact.

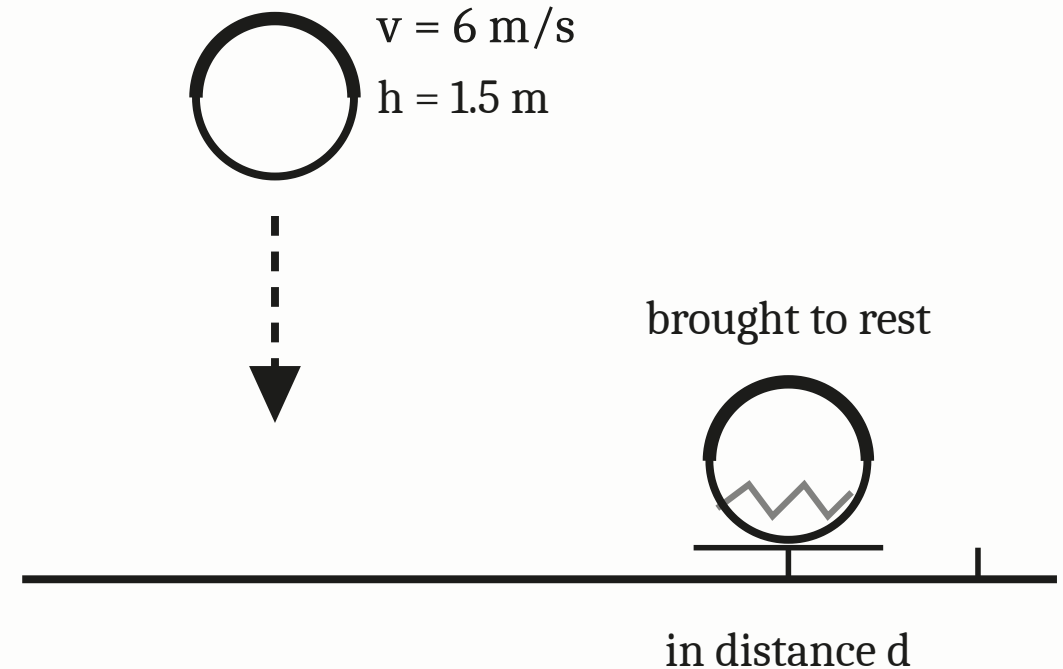
- (a) How much energy must be dissipated?
- (b) What average force does that take if the head stops in **0.5 cm** (no helmet) against **3 cm** (a typical helmet)?

Given:

$$m_{\text{head}} = 5.7 \text{ kg}$$

$$v = 6 \text{ m/s}, h = 1.5 \text{ m}$$

$$d = 0.005 \text{ m, then } 0.03 \text{ m}$$



The crumple zone is a distance, and $W = F \cdot d$

Problem 9 — solution

Six times the distance, one sixth of the force

(a) kinetic plus the potential it still has to lose

$$E = \frac{1}{2}mv^2 + mgh$$

$$E = \frac{1}{2}(5.7)(6)^2 + (5.7)(9.81)(1.5)$$

$$E = 102 + 84 \approx \mathbf{186 \text{ J}}$$

(b) that energy is dissipated by a force over a distance

$$F = E / d$$

$$\text{no helmet: } 186 / 0.005 \approx \mathbf{37,200 \text{ N}}$$

$$\text{helmet: } 186 / 0.03 \approx \mathbf{6,200 \text{ N}}$$

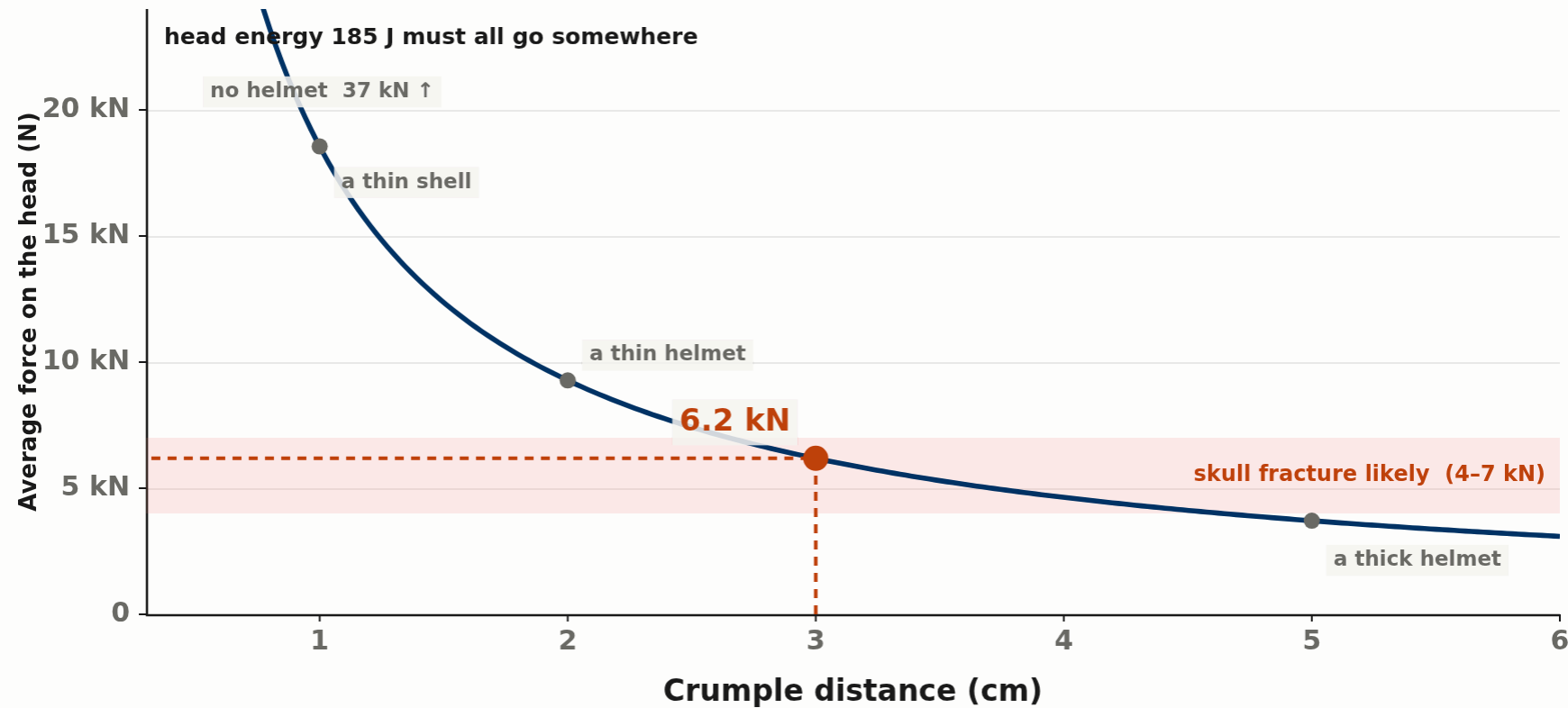
Skull fracture becomes likely somewhere around **4–7 kN**. Without a helmet the impact is roughly **six times** over that threshold; with one it is at its lower edge.

The helmet removed **none** of the 186 J. It only spread it over six times the distance.

- This is the same reasoning as a car crumple zone, an airbag, a crash mat and flexing your knees on landing
- Every one of them is buying **distance**, because that is the only term in $W = F \cdot d$ you can control once the energy is fixed

Problem 9 — the whole curve

Change the speed, the height or the crumple distance



head 5.7 kg · KE 102 J + PE 83 J = **185 J**, stopped in 3.0 cm → **6.2 kN** · a helmet removes none of the energy — it only spends it over a longer distance, and $W = F \cdot d$ does the rest

no helmet

thin

typical

thick

Part 03

Energy in the body

Generation and absorption, joint power, and the springs

Muscles generate and absorb energy

The same tissue is a motor, a brake and a spring

GENERATION — a source

- Concentric: the muscle shortens while pulling
- Chemical energy becomes mechanical energy
- **Positive** work, **positive** power
- Pushing off, climbing, accelerating

ABSORPTION — a sink

- Eccentric: the muscle lengthens while pulling
- Mechanical energy becomes heat — or tendon strain
- **Negative** work, **negative** power
- Landing, braking, walking downhill



Generation on the way up, absorption on the way down

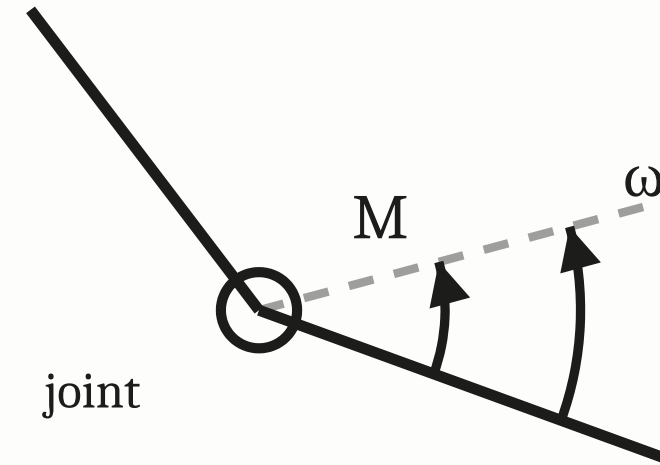
Absorbed energy has two possible fates. It can be **dissipated** as heat and lost, or **stored** elastically in

The rotational form, applied one joint at a time

$$P = M \cdot \omega$$

- **M** is the net joint moment, **ω** the joint angular velocity
- Both are **signed**, and so is the product
- Same sign → **generation**; opposite signs → **absorption**
- Integrate joint power over time and you get the work done at that joint

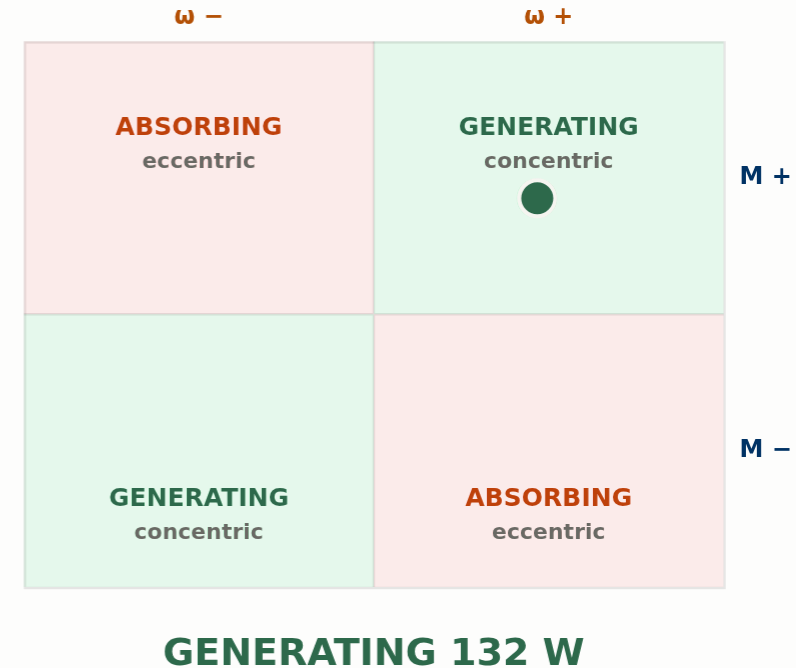
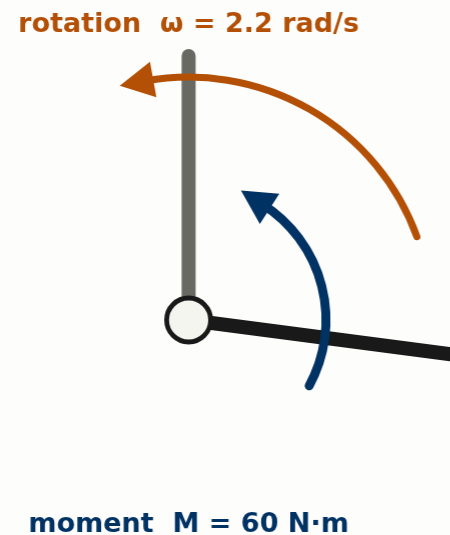
This is how inverse dynamics answers "which joint did the work?" — and the answer in a vertical jump is mostly hip and knee, with the ankle contributing late and briefly.



M and ω in the same sense → the joint is generating

Joint power

Four quadrants — set the moment and the angular velocity



$P = M \cdot \omega = 60 \times 2.2 = \mathbf{132 \text{ W}}$ · moment and rotation agree — the muscle is shortening and putting energy in

Joint
moment
 M

60
 $\text{N}\cdot\text{m}$

Angular
velocity ω

2.2
 rad/s

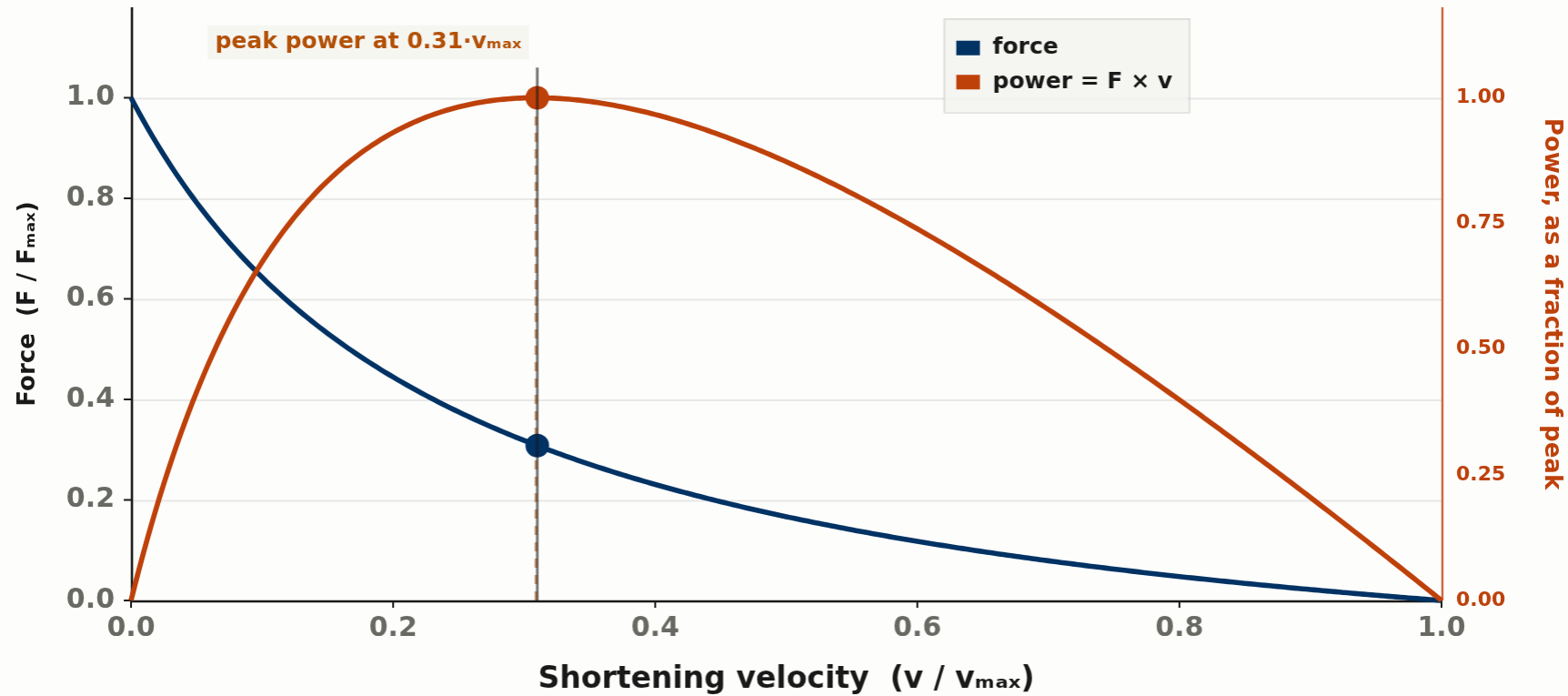
Peak power is at neither end of the range

- **Isometric** ($v = 0$): maximum force, and no power at all
- **$v_{\max}$** : no force, and again no power
- Peak power is at about **$0.3 \cdot v_{\max}$**
- The force there is only about **30%** of maximum

Power is a product, so it is zero whenever either term is zero. The maximum has to lie somewhere in between — and it is much closer to the fast end than most people guess. Olympic lifts and plyometrics are aimed at exactly that region.

Muscle force, velocity and power

Drag along the curve and watch the product



at $v = 0.31 \cdot v_{\max} \rightarrow F = 0.31 \cdot F_{\max} \cdot P = 1.00$ of peak · both terms matter — the product is what training targets

isometric ($v = 0$)

peak power

$v_{\max}$

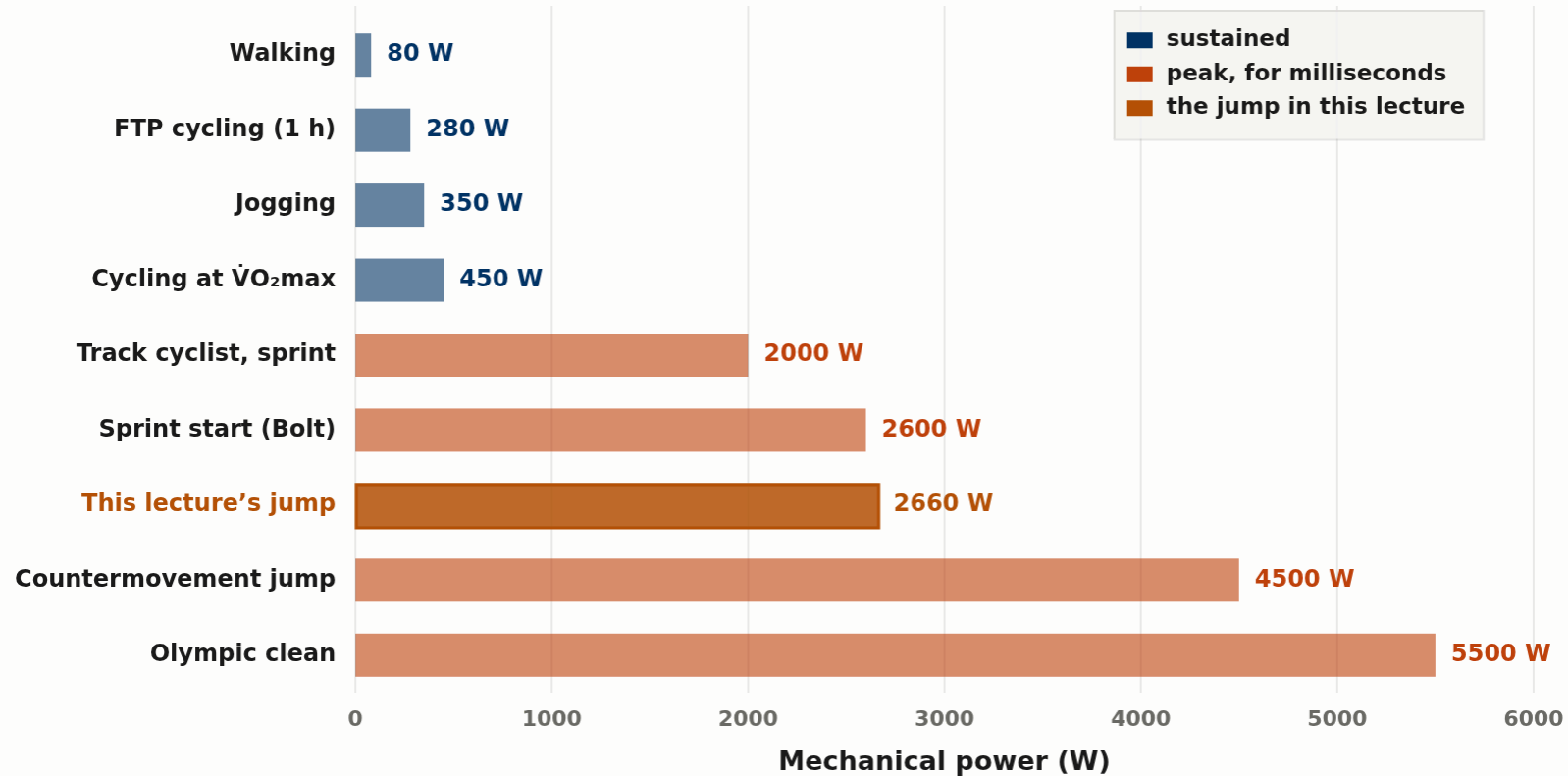
Shortening
velocity



$0.31 \cdot v_{\max}$

Peak power across activities

Explosive movements produce ten to fifty times steady-state power



Peak values last milliseconds — they set sprint starts, jump heights and throws. · a cyclist's watts are measured **at the pedal**; a jumper's are the **whole body**, so the two are not the same measurement

linear

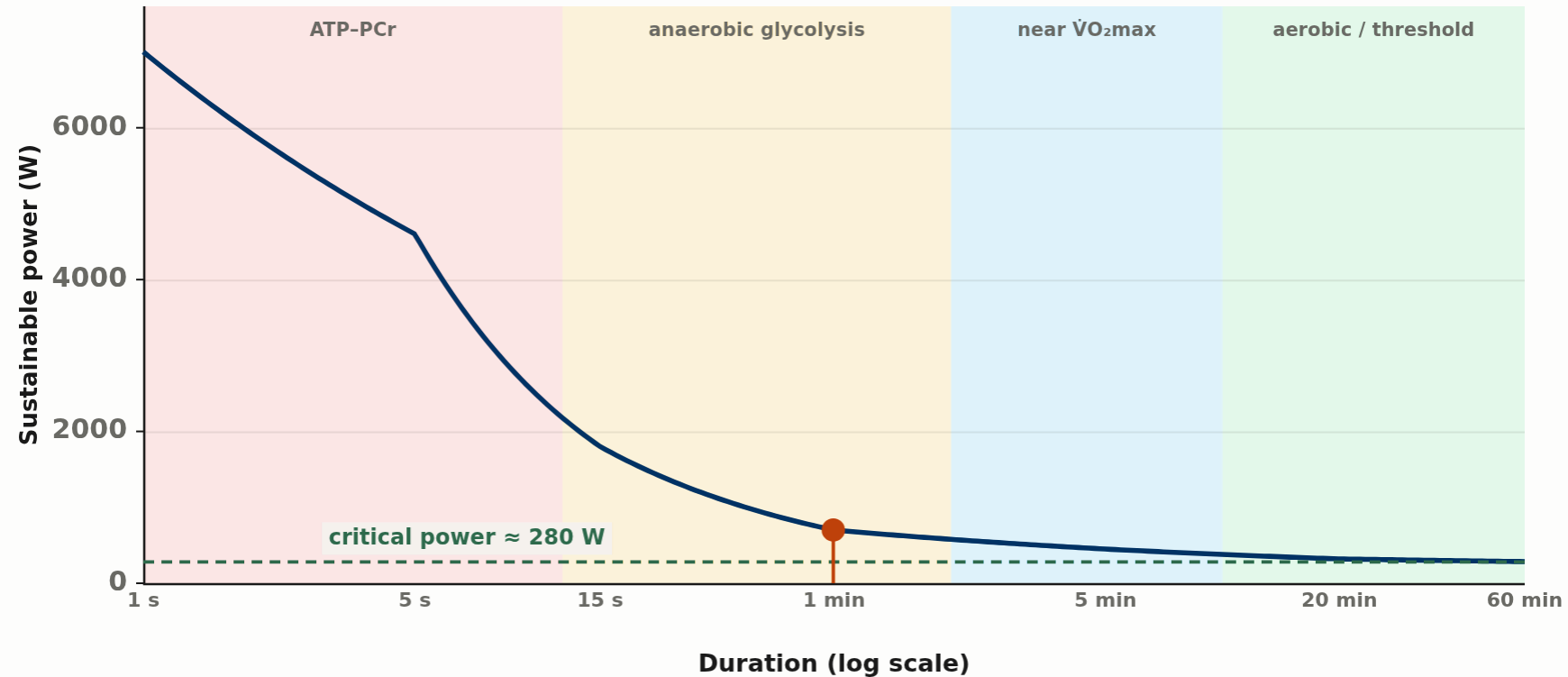
log

by size

sustained first

The power–duration curve

What you can hold falls sharply with how long you have to hold it



for **60 s** → about **702 W** · anaerobic glycolysis · 2.5× critical power

5 s sprint

1 min

5 min

60 min

Duration



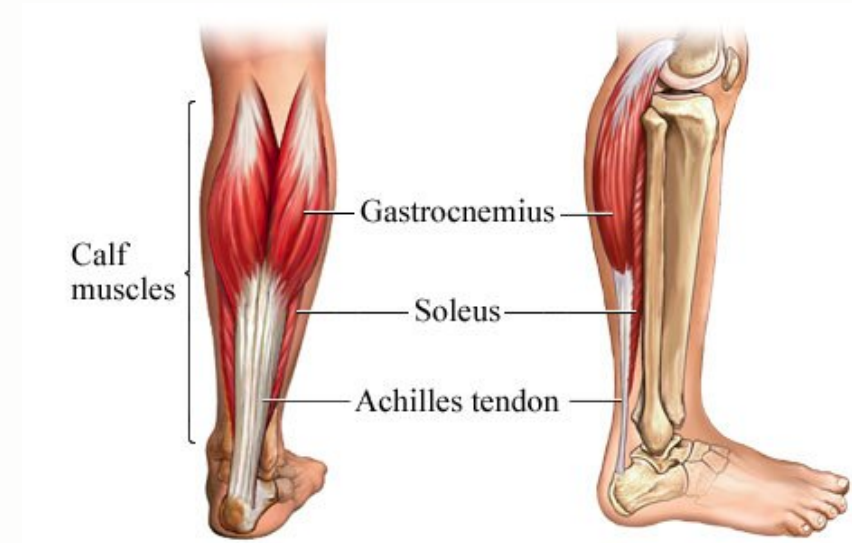
60 s

The stretch–shortening cycle

Tendons store and return energy on every step

- A muscle stretched immediately before it shortens produces **more force and more work** than the same muscle shortening from rest
- The Achilles returns roughly **35–50% of the mechanical energy a running step needs** — energy the muscles never have to produce
- Less muscle work means a lower metabolic cost: running is about 50% more economical than a muscle-only model predicts
- The same mechanism powers countermovement jumps and hopping

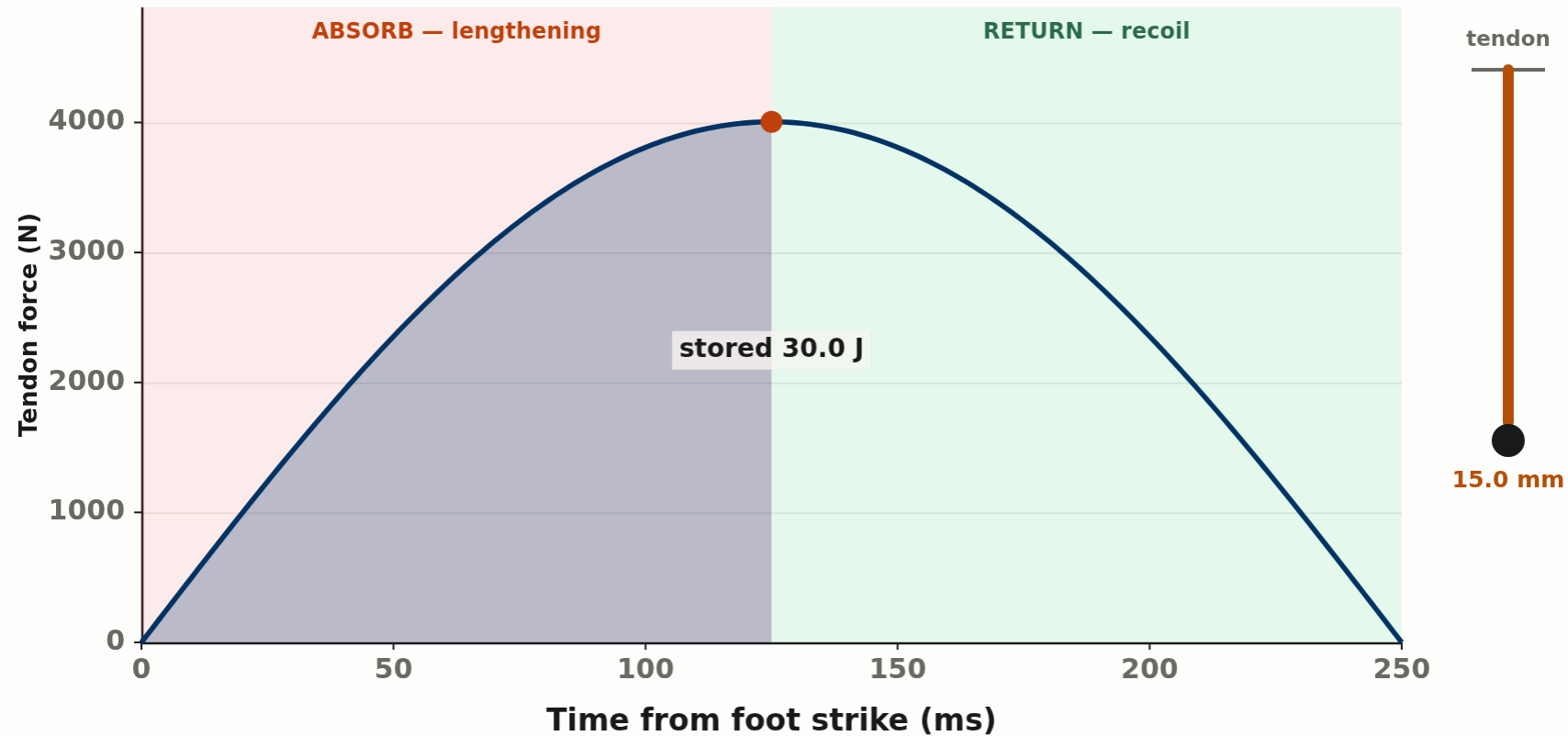
Two percentages that get conflated constantly: the tendon **returns about 90–93% of what it stores** (the rest is hysteresis, lost as heat), and what it stores



Gastrocnemius and soleus in series with one long tendon

The stretch–shortening cycle

One running stance, absorbed and returned



peak stretch **15 mm** stores **30.0 J**; the tendon gives back **27.9 J** (93%) · about 40% of the ≈ 70 J a running step needs — energy the muscles never had to make



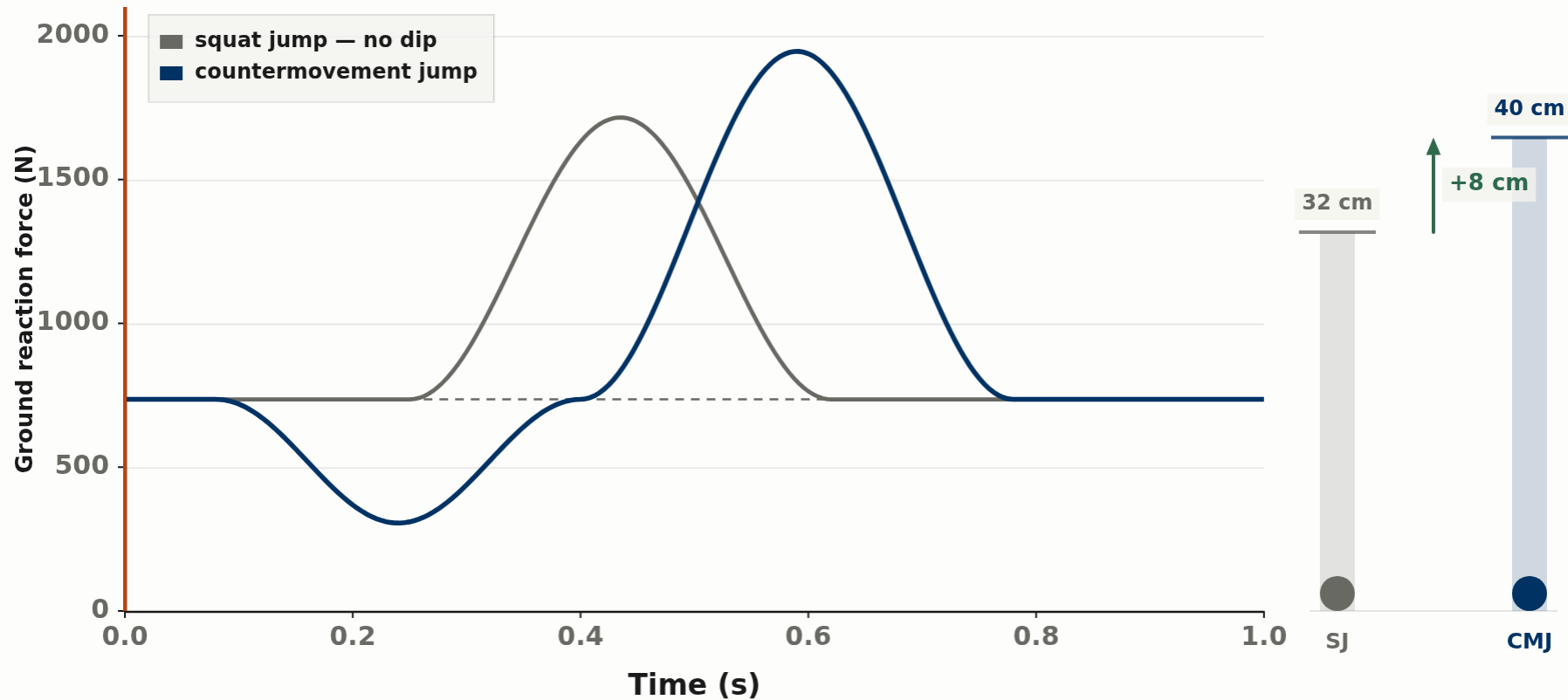
Through the
stance



125 ms

Countermovement against squat jump

Why the dip is worth 8 cm



squat jump **32 cm** · countermovement jump **40 cm** — **+8 cm** · elastic return, force already built before shortening, and a stretch reflex — all three, not one

▶ Play

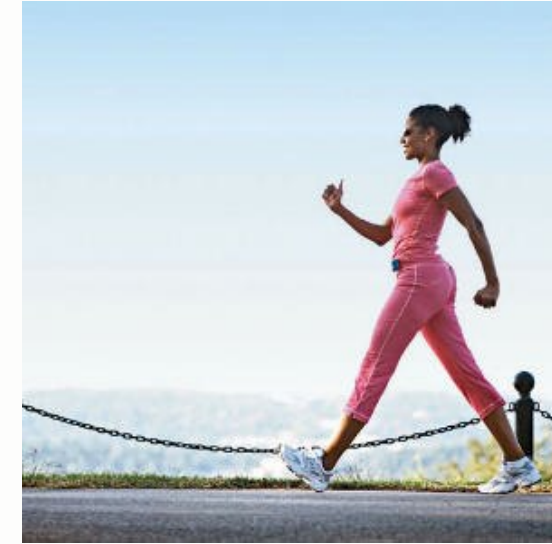
Run both jumps 0.00 s

Walking is a pendulum

The other way the body saves energy — and it needs no tendon at all

- In walking the centre of mass **vaults over** the stance leg, rising and falling about 4 cm each step
- It is **slowest at the top** and fastest at the bottom — so kinetic and potential energy are **out of phase**
- Each is largely supplied by the other, so the muscles have to replace only the difference
- This **inverted pendulum** exchange recovers up to about 70% of the energy a step would otherwise cost

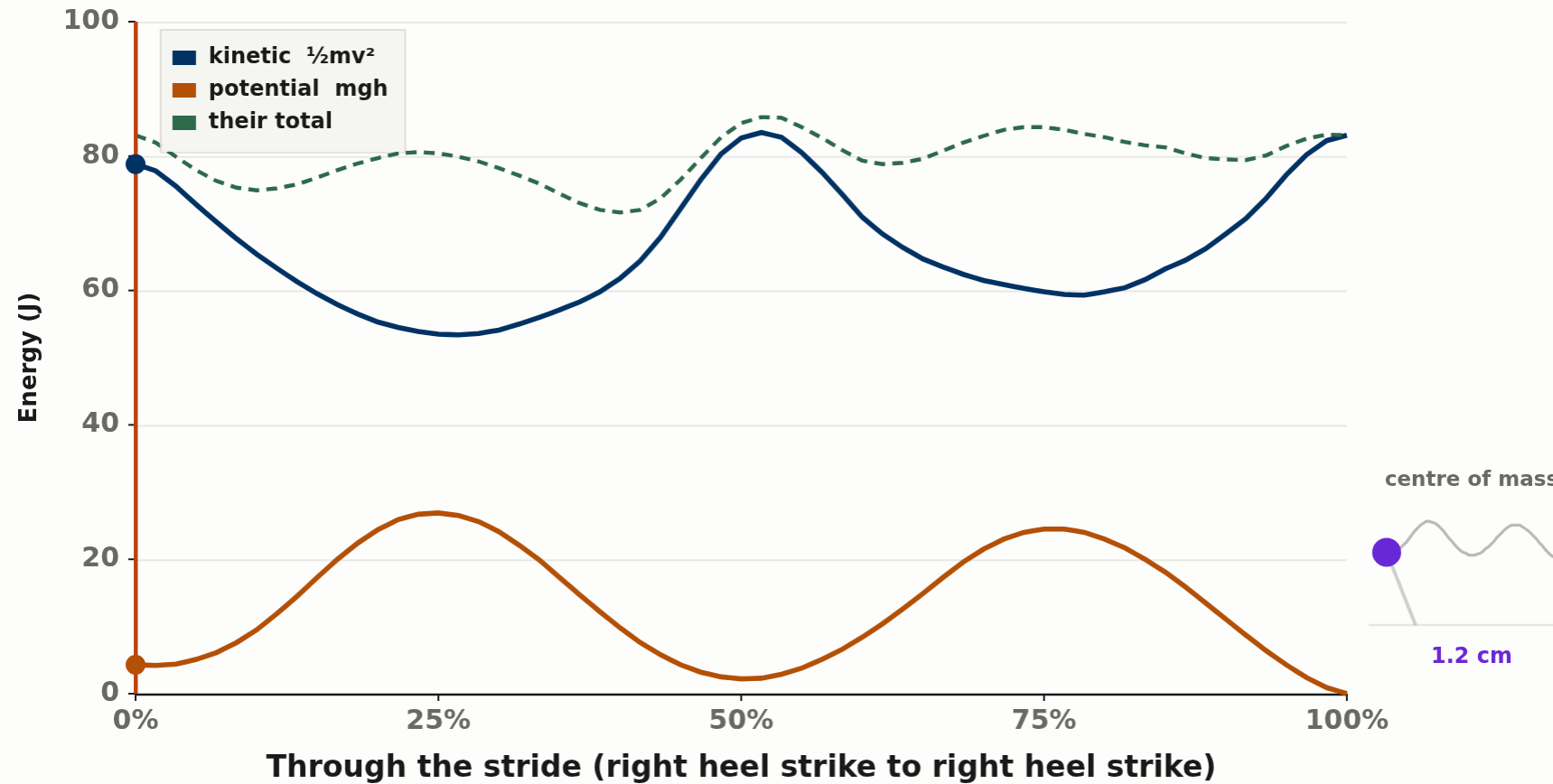
Running works the opposite way: there KE and PE are **in phase** — both lowest at mid-stance — so pendulum exchange is impossible and the saving has to come from the tendons instead.



Vaulting over a straight stance leg

Walking is a pendulum

Measured gait — kinetic and potential, out of phase



KE 79 J · PE 4 J · total 83 J · they run almost exactly out of phase ($r = -0.92$), and 71% of the energy is recovered rather than paid for again

▶ Play

Through the
stride



0%

both + total

kinetic

potential

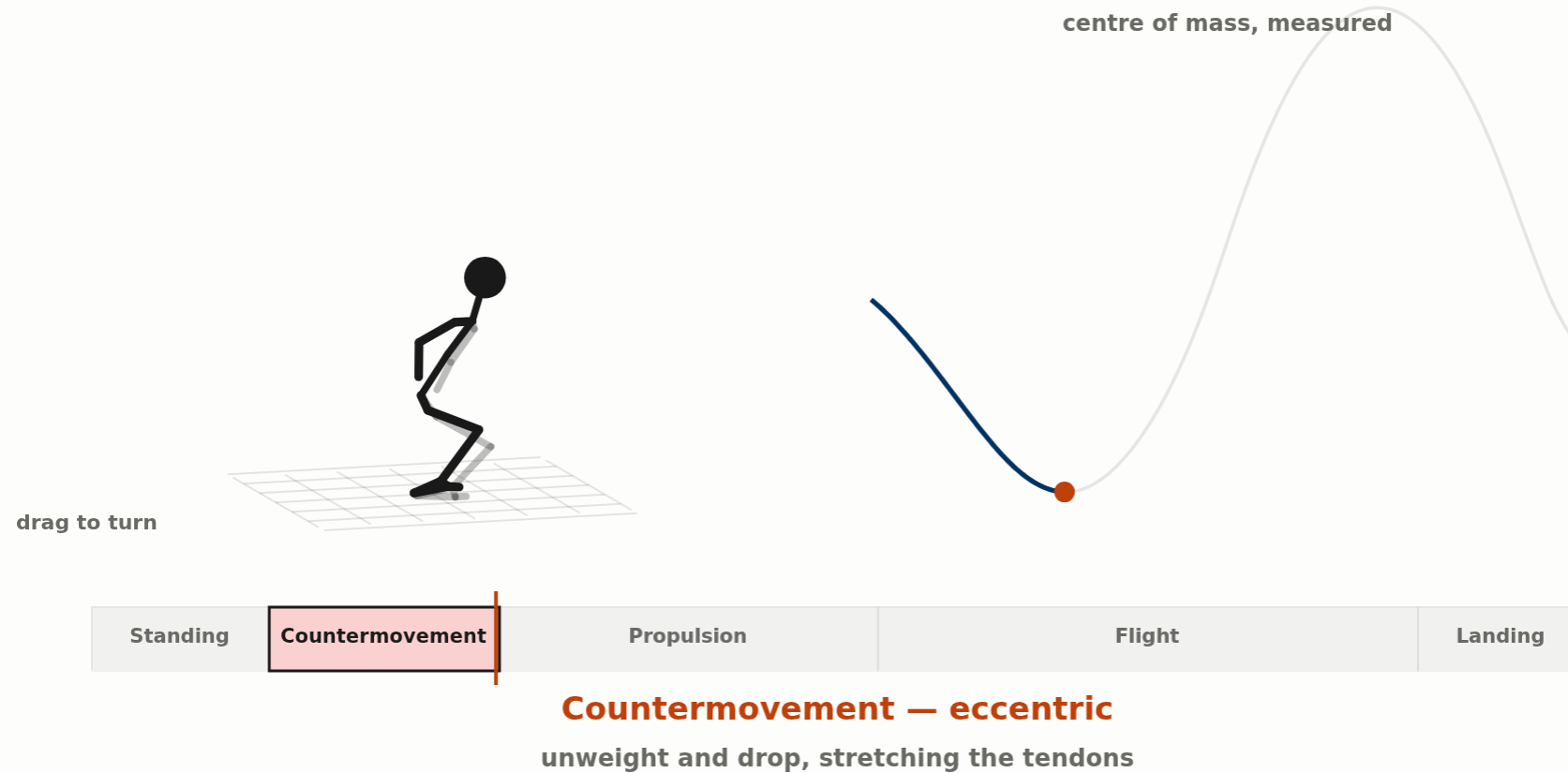
Part 04

One measured jump

The whole lecture read off a single trial

Phases of a countermovement jump

Each phase has its own work and power signature — drag the figure to turn it



Countermovement (eccentric) · unweight and drop, stretching the tendons · CoM -0.34 m from standing · SUSU-46, measured



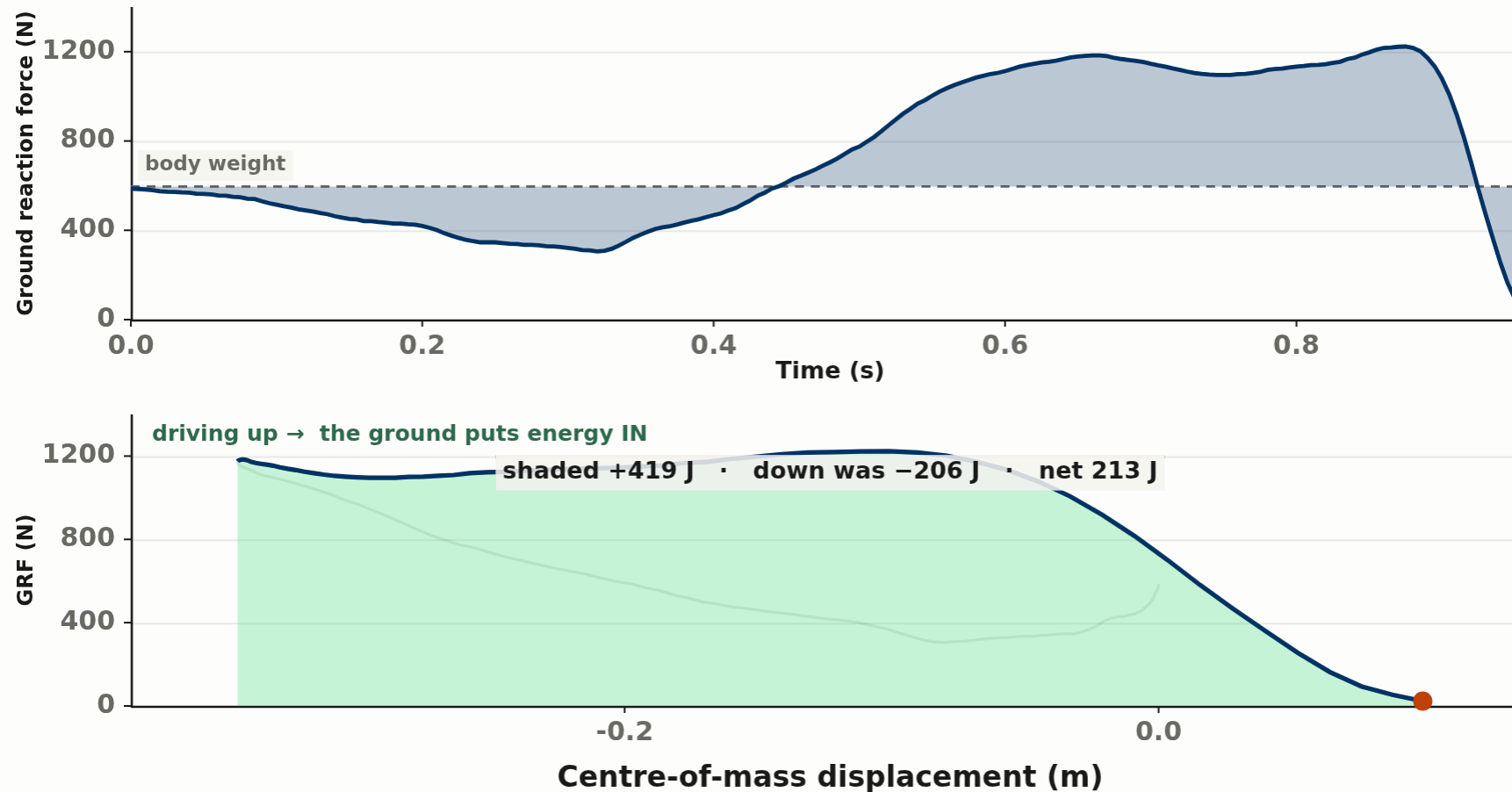
Through the
jump



0.64 s

Work in a vertical jump

The same trial, integrated against displacement instead of time



GRF work so far **213 J** = ΔKE **154 J** + ΔPE **59 J** · $v = 2.25$ m/s · CoM 0.10 m · measured, SUSU-46

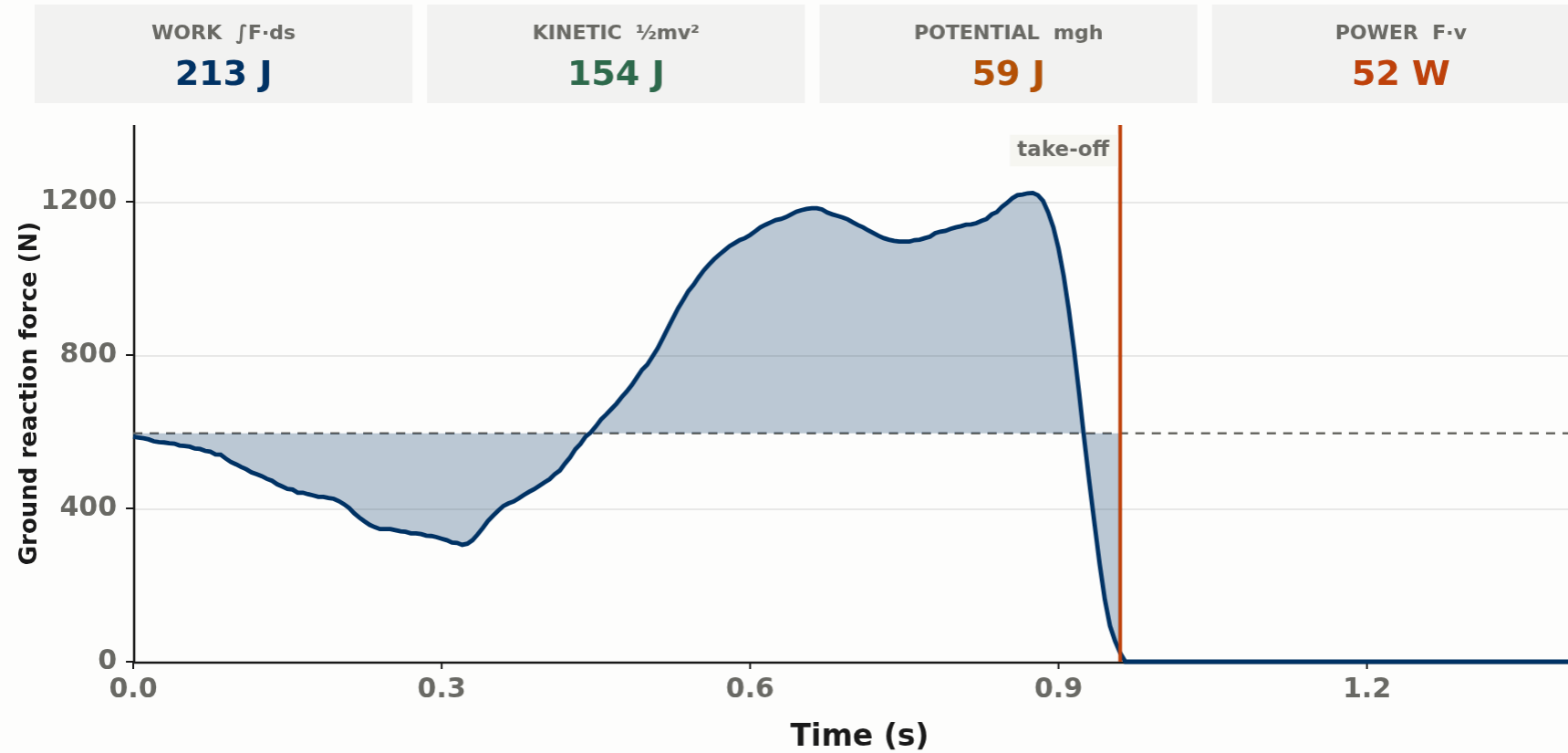


Through the
jump

0.96 s

One movement, four quantities

Work, energy and power are four readings off one force record



one measured jump, four quantities · $v = 2.25 \text{ m/s}$ · $W = \Delta KE + \Delta PE$ at every instant, which is the whole lecture in one line

Through the
jump

0.96 s

The same framework, different dominant terms

Sprinting	$W_{\text{net}} = \Delta KE$	Concentric work accelerates the runner; the stretch–shortening cycle recycles elastic energy each step
Vertical jump	$W_{\text{push-off}} = \Delta(KE + U_g)$	Push-off work becomes take-off velocity, then height — with about 25% added by the countermovement
Walking	$KE \leftrightarrow U_g$	Inverted-pendulum exchange recovers up to 70%; the muscles supply only the shortfall
Landing	$W_{\text{neg}} = -\Delta KE$	Eccentric absorption over the longest distance the joints allow — the helmet problem, done with knees
Cycling time trial	$P = F \cdot v$	Sustainable power at the pedal sets the pace, and the power–duration curve sets the ceiling
Throwing	$W = \Delta KE_{\text{implement}}$	Work is transferred down the kinetic chain and ends as kinetic energy in the hand and the implement

WORK

$$W = F \cdot d \cdot \cos \theta$$

The transfer of energy by a force acting through a distance. Positive when the muscle shortens, negative when it lengthens, zero when nothing moves.

ENERGY

$$KE + U_g + U_e$$

The capacity to do work, in three forms that simply add. Conserved when only gravity and springs act; $W_{\text{net}} = \Delta KE$ otherwise.

POWER

$$P = W/t = F \cdot v = M \cdot \omega$$

The rate of transfer. Positive at a joint means generation, negative means absorption, and muscle power peaks near $0.3 \cdot v_{\text{max}}$.

If you remember one thing: **energy is the currency, work is the transaction, power is the rate.** Every problem in this lecture asked where a fixed number of joules went.

EPHE 341 · BIOMECHANICS

Questions?

Next up: angular kinematics · Further
reading: McGinnis Chpt 4